Sufficient Statistics for Economic Mobility: When Do Measures Agree?*

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Abstract

Different empirical measures of intergenerational mobility—elasticities, rank–rank slopes, transition matrices—can rank the same places or periods differently. I show that concordance guarantees agreement: when one economy is everywhere more rank-preserving than another, standard measures rank them identically, and for transition probabilities the condition is also necessary. Changes in any measure decompose into positional dependence and distributional components. In a Becker–Tomes–Loury model, concordance depends on which families' children gain most, not on average skill gains or the return to skill. Testing concordance in twentieth-century U.S. data, I find long-run gains in mobility for every rank–based measure, while the elasticity's post-1950 rise reflects growing inequality. The test rejects any ordering only for cohorts born in the 1950s and 1960s, where mobility falls across most of the distribution but rises at the top—the cohorts on which existing estimates differ.

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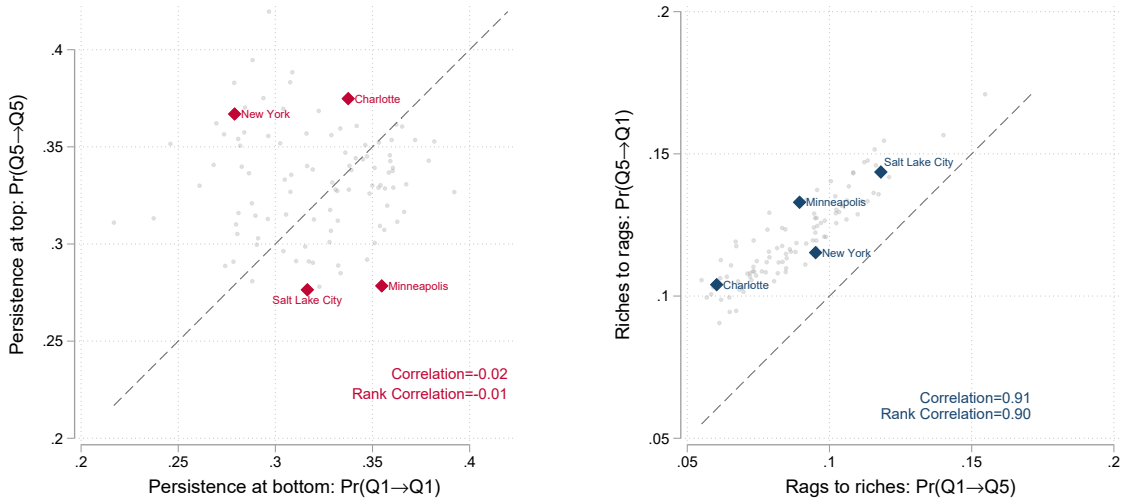
1. Introduction

Has intergenerational mobility in the United States declined? Answering this requires choosing among a variety of mobility statistics—elasticities, rank–rank slopes, transition matrices, or axiomatic indices—each of which provide different summaries of the joint distribution of parent and child outcomes.[1] Yet the answer often depends on which statistic, and which data, are used. or cohorts born around mid-century, Davis and Mazumder (2024) find the rank–rank slope rose by roughly half. Jácome, Kuziemko, and Naidu (2025) find little change in the same slope, even as the elasticity climbs, and post-1950 trends are sensitive to the measure used.

Rankings of places show the same sensitivity (Blanden 2013). Figure 1 plots transition probabilities across the 100 largest U.S. commuting zones constructed from Chetty et al. (2014a)'s published matrices. Persistence at the bottom and persistence at the top are essentially uncorrelated, disagreeing about which of two places is more mobile for half of all pairs. Whether Salt Lake City is more mobile than New York, for instance, depends on which tail is used. Even rags-to-riches and riches-to-rags probabilities, which are strongly correlated, disagree for roughly one pair in eight. In other contexts, these measures, or summary statistics, produce consistent rankings of mobility (e.g., Katz and Krueger 2017, Berman 2022, Deutscher and Mazumder 2023). This raises two fundamental questions: when should we expect standard mobility measures to deliver the same ranking of economies, and when should disagreement be interpreted as substantive or indicative of underlying economic mechanisms?

This paper proposes a framework to organize these agreements and disagreements. When the parent–child joint distribution in one economy is everywhere more rank-preserving—i.e., when parent and child ranks become more closely aligned throughout the distribution—positional mobility measures provide the same ordering of mobility. Measures are then *sufficient statistics* for overall mobility and become substitutes—

[1] A large body of research uses these measures to document differences across countries (e.g., Corak 2013; Jäntti et al. 2006), regions (e.g., Chetty et al. 2014a; Corak 2020), and over historical periods (e.g., Feigenbaum 2018; Collins and Wanamaker 2022). Researchers use mobility measures to study income across and within generations (e.g., Solon 1992; Chetty et al. 2014b), wealth (e.g., Boserup, Kopczuk, and Kreiner 2018; Fagereng et al. 2020; Fagereng, Mogstad, and Rønning 2021; Audoly et al. 2024), consumption (e.g., Jappelli and Pistaferri 2006), health (e.g., Halliday 2023; Black et al. 2024), education (e.g., Black and Devereux 2011), socio-emotional skills (Attanasio, De Paula, and Toppeta 2025), or occupational prestige (Mazumder and Acosta 2015; Olivetti and Paserman 2015; Song et al. 2020; Haeck and Laliberté 2025). They are used in positive and normative analysis (e.g., Shorrocks 1978; Atkinson and Bourguignon 1982), evaluating policy impacts (e.g., Chetty and Hendren 2018a,b), and as validation or targets for structural economic models (e.g., Abbott et al. 2019; Bolt et al. 2025; Daruich 2018)



(A) Tail Persistence (B) Tail Mobility

FIGURE 1. Tail Mobility Measures Across Commuting Zones (Chetty et al. 2014a)

Notes: This uses only the 100 largest commuting zones in Chetty et al. (2014a) by 2000 population, within-CZ copulas reconstructed from the published national-quintile matrices (see Appendix A for data construction details), and labelled CZs chosen for illustration. Panel (A) reports transition probabilities using quintiles of the within-CZ distribution for the probability of remaining in the bottom and top quintiles across generations. Panel (B) reports probabilities of transitioning from the bottom quintile to the top quintile and vice versa.

including intergenerational elasticities when marginal distributions are unchanged. For the full family of threshold transition probabilities, the characterisation is exact as they are ordered in the same direction at every pair of parent and child thresholds if and only if the economies satisfy the rank–preserving property. This observation explains why studies that report multiple measures, either as distinct objects of interest or robustness checks, typically find broadly consistent results.

This substitutability has immediate practical implications. Researchers have advanced different arguments for preferring elasticities, rank–rank slopes, or transition matrices on grounds of interpretability, robustness, or transparency.[2] When one economy is everywhere more rank-preserving than another, all measures agree on which economy is more mobile, and researchers can choose based on data availability

[2]The intergenerational elasticity is widely used because it is interpretable, model-consistent, and comparable to an established evidence base (Black and Devereux 2011), though it is sensitive to life-cycle bias (Haider and Solon 2006; Nybom and Stuhler 2017). Rank–rank slopes are robust and unit-free (Chetty et al. 2014a). Transition matrices are transparent (Jäntti and Jenkins 2015) with minimal data requirements, but depend on discretization (Cowell and Flachaire 2018).

or inference concerns. Nor is this an arbitrary statistical restriction. Bounding exercises for absolute mobility impose the "*intuitive requirement that children from higher-income families are less likely to have lower incomes*" (Chetty et al. 2017, p. 401; see also Berman 2022; Manduca et al. 2024), and work using transition matrices treats substantial anti-diagonal mass—high-income parents systematically producing low-income children and vice versa—as implausible (Shorrocks 1978; Jäntti and Jenkins 2015). I formalise these restrictions as a uniform, rank-preserving shift in the joint distribution of parent and child ranks.

Substitutability follows from a single dependence order on the parent–child copula, known as concordance (Tchen 1980). The copula captures the dependence between parent and child ranks, independent of the marginal income distributions in each generation. The concordance order is testable through pointwise inequality restrictions on the copula, grounded in standard models of human capital investment, and orders many parametric families used in applied work. Its implications are also robust to measurement error and to proxies, such as education or occupation, in place of lifetime income provided measurement is common across the economies compared.[3] The role of concordance parallels that of Lorenz dominance in inequality measurement. Just as Pigou–Dalton transfers generate an ordering under which all transfer-sensitive inequality measures agree (Atkinson 1970), concordance-increasing exchanges generate an ordering under which standard exchange-mobility measures agree.

Concordance-increasing shifts arise naturally in Becker and Tomes (1979)–Loury (1981) models under standard economic forces (Section 5). Whether a reform increases the average level of skills in the economy is irrelevant to characterising positional, or exchange, mobility. Instead, the "*measures-as-substitutes*" result depends on the differential incidence of skill changes. When skills rise proportionally more for lower income households mobility increases and concordance decreases. Not all economically relevant changes take this form, however. Policies or shocks that improve outcomes for some parts of the distribution while weakening the parent–child link elsewhere—such as targeted interventions for disadvantaged families that coincide with increased sorting among elites—can cause the concordance order to fail. A higher return to skill, by contrast, stretches the income distribution without reordering children, leaving the copula unchanged. Rank–based measures track who gains skill, while the IGE also tracks how skill is priced.

[3]When concordance orders economies, measures are ordinal substitutes: they agree on which economy is more mobile even though their values and units differ. The term "sufficient statistic" is therefore not used in the Fisher-information sense.

The framework can also be applied to understand why measures diverge. First, any change in a mobility measure decomposes into a component due to changing dependence (the copula) and a distributional component due to changing marginal distributions. Thus, even when dependence changes uniformly, measures may reflect changes in inequality, dispersion, location, skewness, or other features of the income distribution. Rank–based measures—rank–rank slopes, transition matrices, the Shorrocks index—are invariant to the marginal distributions, since ranks are uniformly distributed regardless of the underlying income distribution. Second, measures need not agree, because they weight different regions of the joint distribution. I show precisely which regions each emphasises and that even the simplest measures can conflict: whenever concordance fails, some threshold transition probabilities rank the economies in opposite directions. I therefore propose a reporting protocol based on local rank–rank slopes and transition probabilities, and discuss principled completions of the partial order when a single ranking is required.

Section 8 uses the framework to assess whether U.S. mobility increased over the 20th century and declined after mid-century, a question on which existing estimates differ (Davis and Mazumder 2024; Jácome, Kuziemko, and Naidu 2025). Positional mobility never reverted to early–century levels. Every later cohort is consistent with weaker rank dependence than the 1910s cohort, and the post-1950 rise in the IGE primarily reflects rising income dispersion rather than stronger rank persistence. Adjacent cohorts are more nuanced. The ordering between the 1940s and 1950s cohorts is driven by rising persistence at the top of the child distribution. Between the 1950s and 1960s the copulas are unordered. Mobility falls across most of the distribution, but rises at the very top. Whether mobility declined for this cohort depends on which part of the distribution a measure emphasises—even the IGE and rank–rank slope disagree with income distributions held hypothetically fixed. Thus, results are not robust to researcher choice.

1.1. Related Literature

A large body of work proposes and estimates mobility measures (surveys include Black and Devereux 2011, Deutscher and Mazumder 2023). These studies ask which measure best captures underlying mobility. This paper instead asks, under what conditions do these measures deliver the same ordering of economies?

The closest theoretical work is D'Agostino and Dardanoni (2009), who axiomatise concordance as a measure of rank mobility in a discrete transition-matrix setting and

cardinalise by scaling against a maximal benchmark. Atkinson and Bourguignon (1982) study how dependence shapes utilitarian welfare rankings in multidimensional inequality, using restrictions on the joint distribution of economic status that are formally similar to those employed here. This paper differs in three respects. First, rather than proposing a new index, I show that concordance is a common condition for measures already used in applied work and in environments with proxies or non-classical measurement error. Several of these sufficiency results follow from classical dependence orderings (Lehmann 1966; Tchen 1980), and I show where they require additional structure (Assumption 1). Second, concordance is also necessary: uniform agreement of threshold transition probabilities holds only under concordance, and its failure produces ranking reversals even on standard quintile and decile grids. In the U.S. data, the 1950s and 1960s copulas cross, yet the measures researchers typically report move together; agreement among a handful of measures is therefore not evidence of robustness. I show how to test for and locate such violations. Third, in a Becker–Tomes–Loury model I show that what determines concordance is the incidence of skill gains across parental resources, not their average level or the price of skill.

I study the properties of *estimands* rather than estimators, complementing work on inference, identification, and bias in mobility estimates (e.g., Chetverikov and Wilhelm 2023; Kitagawa, Nybom, and Stuhler 2018; Collado, Ortuño-Ortín, and Stuhler 2023; Espín-Sánchez, Ferrie, and Vickers 2023; Haider and Solon 2006; Nybom and Stuhler 2017). The conditions I provide apply regardless of how population parameters are estimated, and concordance orderings survive measurement error common across economies (Section 3).

This paper studies positional (exchange) mobility and is silent on absolute mobility measures such as the fraction of children exceeding the median income of the parent generation (Katz and Krueger 2017). Ray and Genicot (2023) show that panel-independent mobility measures can be constructed when welfare depends only on growth progressivity. The concordance framework complements this work: while Ray and Genicot characterise planner preferences over upward mobility, I characterise the dependence structure determining relative positional mobility. Some absolute mobility bounds (Chetty et al. 2017; Berman 2022; Manduca et al. 2024) and parametric copula models (e.g., Hu and Zhou 2025; Cai et al. 2026) impose copula restrictions; concordance clarifies when those restrictions are empirically plausible. Cai et al. (2026) apply the concordance test developed here. Coleman (2025) studies copula-based measures of dependence and exchange mobility for discrete outcomes, whereas I

characterise when such measures agree in ranking economies.[4]

This paper relates to work studying sorting through concordance properties. Gola (2021), Anderson and Smith (2024), and Boerma et al. (2023) derive concordance orderings as equilibrium outcomes of assignment problems and use them to measure sorting in the labour market. Chiappori et al. (2025) axiomatise an odds-ratio index for assortative mating that is both necessary and sufficient for concordance in discrete contingency tables; their normalised-trace index is a direct transformation of the Shorrocks (1978) mobility measure. I apply similar tools to a different question, but the implication that concordance makes measure choice redundant carries over directly to assortative mating and matching in marriage and labour markets, where sorting can strengthen in some regions while weakening in others (Eika, Mogstad, and Zafar 2019; Chiappori et al. 2025). Moreover, I link axiomatic properties of mobility measures to optimising behaviour (see also Chiappori et al. 2025) giving sufficient conditions under which a Becker–Tomes–Loury model generates concordance-ordered economies.

2. Framework: Copulas and Concordance

This paper considers the following setting. A researcher compares mobility across economies through the joint distribution of parent–child (or period-to-period) outcomes.

Let (Y^K, Y^P) denote child and parent outcomes with joint CDF $F(Y^K, Y^P)$ and marginals $F_K(Y^K)$ and $F_P(Y^P)$. The key insight is that positional mobility—how children's positions relate to parents' positions—is characterised by the rank dependence structure between these outcomes, not their levels or dispersion.

I denote the *position* of an individual by their rank in their generation's marginal distribution. In the continuous case ranks, $R^K = F_K(Y^K)$ and $R^P = F_P(Y^P)$, are uniformly distributed on $[0, 1]$ and map from incomes to the unit interval. I break ties uniformly and independently at random within each mass point (see Appendix B for a definition).

This dependence structure is captured by the copula. By Sklar's Theorem (1959), the joint distribution can be decomposed as

$$F(Y^K, Y^P) = C\left(F_K(Y^K), F_P(Y^P)\right), \tag{1}$$

[4] I focus on bivariate parent–child distributions. Orthant orders (Shaked and Shanthikumar 2007) generalise the concordance order to multivariate settings and Audoly et al. (2024) adopt hierarchical clustering to summarise high-dimensional wealth and income histories. The bivariate focus isolates the core dependence structure in most empirical work and simplifies exposition.

where $C : [0,1]^2 \to [0,1]$ is the copula or joint distribution of ranks. The *copula alone* encodes all rank-dependence between generations, independently of marginal distributions.[5]

While Sklar's Theorem (1959) applies to arbitrary distributions, including those with mass points, the copula is not unique when the marginals are non-continuous. For distributions with mass points, the random tie-breaking convention above selects the checkerboard (or multilinear) extension, under which

$$C(u, v) = \Pr\left(R^K \leq u\,,\ R^P \leq v\right). \tag{2}$$

This extension is uniquely defined by multilinear interpolation and preserves concordance comparisons between joint distributions with common marginals (Genest and Nešlehová 2007). Appendix B provides the formal construction and discusses alternative extensions. This convention allows the main concordance framework to accommodate economically important mass points arising, for example, from bunching, censoring, bottom-coding, or non-participation.

To isolate positional or rank dependence, the main analysis will fix marginal distributions and focus on changes in the copula. This corresponds to a thought experiment in which the U.S. marginal income distributions remain constant, but the parent–child rank-dependence structure varies. Under this restriction, each mobility measure becomes an alternative summary of the same copula or how often "high" outcomes for one generation coincide with "high" outcomes for the next.[6] Section 4 shows how relaxing the constant-marginals assumption affects the results.

2.1. The Concordance Order

I begin by illustrating changes in dependence with an example which assumes outcomes take on discrete values for simplicity, then formalize the concordance order.

Figure 2 presents a grid of parent–child quintile transition matrices. Moving rightward increases concordance: probability mass shifts from origin independence (Prais 1955) to diagonal (rank-preserving) cells with the right showing perfect concordance.

The concordance order formalizes the intuitive idea that "large" values of one

[5] For simplicity I abstract from inference (see Chetverikov and Wilhelm 2023).

[6] This property is exploited by Chetty et al. (2017) and Manduca et al. (2024), who impose copula restrictions to bound absolute mobility when panel data are unavailable. I exploit the same logic here by summarizing relative mobility through the copula.

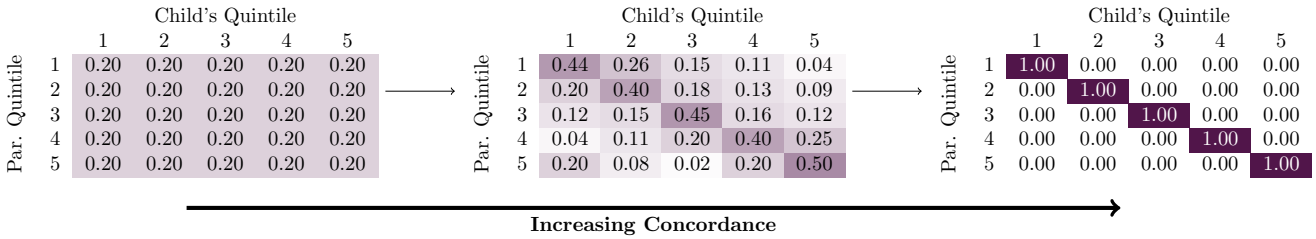


FIGURE 2. Concordance Ordering Among Transition Matrices

Notes: Moving rightward increases concordance—probability mass shifts from off-diagonal (high-parent/low-child or vice versa) to diagonal (similar ranks across generations). Left: independence (no association). Right: perfect rank-preservation (children inherit parental quintiles exactly). Empirical mobility estimates typically fall between these extremes.

random variable tend to coincide with "large" values of another, a form of stochastic dominance for dependence in joint-distributions (e.g., Kirkegaard 2017). In the context of intergenerational mobility copulas, concordance is higher if parental ranks are more likely to be realised with higher child ranks. Importantly, it appeals to an implicit sense of mobility. For example, the bounding exercise in Chetty et al. (2017) assumes the copula satisfies the "*intuitive requirement that children from higher-income families are less likely to have lower incomes*" (pg. 401). Concordance formalises these types of intuitions in a mathematically precise and testable stochastic ordering over copulas:[7]

DEFINITION 1 (Concordance Ordering (Yanagimoto and Okamoto 1969; Tchen 1980)). *Let C^A denote the copula, or joint distribution of ranks, in economy A and C^B denote the copula in economy B. The dependence between outcomes Y^K and Y^P is said to be larger in concordance in economy A than economy B if and only if $C^A(u,v) \geq C^B(u,v)\ \forall\, u,v \in [0,1]$.*

This is denoted $C^A \succeq C^B$ or $A \succeq B$.

Note that higher concordance corresponds to lower mobility, as parent–child ranks become more strongly aligned: $C^A \succeq C^B$ implies less mobility in A than B. Thus, 'increasing concordance' and 'decreasing mobility' are synonymous. Section 8 develops a formal test of these copula inequalities using empirical checkerboard-copula inference and generalized moment selection (Andrews and Soares 2010).

Formally, the concordance order captures two distinct concepts. First, the degree of monotone dependence or the tendency of random variables to cluster around the

[7]Throughout I use superscript A and B to denote primitives and measures in economies A and B, respectively. This ordering is sometimes alternatively known as the PQD ordering or point-wise copula ordering.

graph of any (measurable) monotone function $Y^K = f(Y^P)$ or $Y^P = g(Y^K)$. Second, the direction of monotonicity—whether the functions are monotone increasing or decreasing. The concordance order runs from maximum mobility (perfect rank-reversal between generations, the Fréchet lower bound)[8] to minimum mobility (perfect rank-preservation, the Fréchet upper bound).

Geometric interpretation. Any concordance increase can be decomposed into elementary 'exchanges': pick any rectangle in rank-space and shift probability from corners where parents and children have opposite relative positions (one high, one low) to corners where both are high or both are low[9] The following elementary exchanges define a sequence of copula transformations ordering joint-densities by their concordance (see also Tchen 1980):

DEFINITION 2 (Concordance Increasing Exchanges). *Let C^A be a copula. Pick $0 < u_1 < u_2 < 1$ and $0 < v_1 < v_2 < 1$, and let $\varepsilon > 0$ be small enough that the update below remains a valid copula. Define*

$$C^B(u,v) = C^A(u,v) + \varepsilon\Big(\mathbf{1}_{\{u\geq u_1,\, v\geq v_1\}} + \mathbf{1}_{\{u\geq u_2,\, v\geq v_2\}} - \mathbf{1}_{\{u\geq u_1,\, v\geq v_2\}} - \mathbf{1}_{\{u\geq u_2,\, v\geq v_1\}}\Big).$$

We say C^B is obtained from C^A by a concordance-increasing exchange *on the rectangle $[u_1, u_2] \times [v_1, v_2]$.*

This operation preserves the marginals, $C^B(u,1) = u$ and $C^B(1,v) = v$, and moves probability mass from the two off-diagonal corners of the rectangle to the two diagonal corners. The reverse update with $\varepsilon < 0$ is concordance-decreasing and the operation can be confined to arbitrarily small rectangles ($u_2 - u_1 < \eta$ and $v_2 - v_1 < \eta$, for $\eta > 0$). Any concordance increase can be approximated by a sequence of such exchanges (Tchen 1980). This provides a simple building block for increased concordance.

These exchanges are the mobility analogue of Pigou–Dalton transfers. Just as transfers generate Lorenz dominance, exchanges generate the concordance order,

[8] This is akin to maximal negative rank correlation and generalizes Prais (1955)'s origin independence which corresponds to the independence copula. This is analogous to Negative Assortative Matching as the reverse of Positive Assortative Matching rather than the absence of sorting.

[9] For finite economies with N dynasties, concordance has an equivalent characterization through pairwise exchanges. Consider two dynasties i and j satisfying $(Y_i^K - Y_j^K)(Y_i^P - Y_j^P) < 0$. Swapping their children's outcomes increases concordance while preserving both marginal distributions. Treating such exchanges as reductions in positional mobility provides a natural elementary axiom, analogous to the Pigou–Dalton transfer principle in inequality measurement.

which Section 3 shows is sufficient for exchange-mobility measures to agree. Its failure (Section 6) mirrors crossing Lorenz curves.

Families of Distributions. A large class of multivariate distributions are ordered by their concordance. While the concordance order is nonparametric, many common parametric families are monotonically ordered by concordance as their dependence parameter varies (Joe 2014). This includes bivariate normal, elliptical, and Archimedean copulas (Clayton, Gumbel, Frank) as well as multivariate extensions of the Singh and Maddala (1976) size-distribution for incomes, generalizing Champernowne (1952); Fisk (1961) and Pareto distributions. The concordance order is preserved by increasing marginal transformations. Moreover, if two continuous rank distributions are ordered by concordance, any common finite partition of rank space preserves the corresponding cumulative inequalities at the partition boundaries. Thus, the induced finite transition matrices inherit the concordance ordering in the natural discrete sense. Moreover, the order is preserved under convex combinations and weak limits: mixtures of more-concordant copulas remain more concordant, and pointwise limits of concordance-ordered sequences inherit the order (Shaked and Shanthikumar 2007).

3. Concordance Ordering and Ordering Mobility Measures

Having established concordance as an intuitive dependence concept for mobility, I now show how this orders commonly used exchange mobility measures—from regression-based to axiomatic indices, operating on ranks, logs, or income level—as well as how it applies under fixed rank comparisons, proxies and measurement error. This demonstrates that concordance ensures substitutability. Thus, when it applies, researchers need not choose between competing metrics, as they all provide the same ranking over positional mobility.[10] I build on Deutscher and Mazumder (2023)'s taxonomy and focus on exchange mobility measures that depend solely on the joint-distribution. In applied work Y is most commonly income (e.g., Chetty et al. 2014b), although other measures such as wealth (e.g., Fagereng, Mogstad, and Rønning 2021; Audoly et al. 2024) or health (e.g., Halliday 2023) are also used. These coherence results are presented sequentially, summarized in Table 1 by methodological approach, and formal proofs are in Appendix C.

[10] This is 'sufficient statistic' in the ordinal sense: concordance determines measure-rankings.

TABLE 1. Which Measures Agree Under Concordance? A Practical Reference

	Properties		Concordance Ordering				
	Copula only?	Distribution of what?	Increasing?	Mass points?	Fixed reference ranks?	Common Proxies or measurement error?	Common marginals only?
Regression or Correlation measures							
IGE	No	Log income	Yes	Yes	N/A	Yes	Yes
IGC (Pearson correlation)	No	Log income	Yes	Yes	N/A	Yes	Yes
Rank–rank correlation	Yes	Ranks	Yes	Yes	Yes	Yes	No
CER	Yes	Ranks	Yes	Yes	Yes	Yes	No
Transition matrix measures							
TP	Yes	Ranks	Yes	Yes	Yes	Yes	No
URM/DRM	Yes	Ranks	Yes[a]	Yes	Yes	Yes	No
Axiomatic measures							
Shorrocks	Yes	Ranks	Yes[a]	Yes	Yes	Yes	No
Fields & Ok	No	Income levels or log income	Yes	Yes	N/A	Yes	No[b]
D'Agostino & Dardanoni	Yes	Ranks	Yes	Yes	Yes	Yes	No
Cowell & Flachaire	User choice	Income levels or ranks	Yes	Yes	Yes	Yes	No[b]

Notes: [a] refers to results that additionally require Assumption 1. Most results require only concordance (Definition 1). Two results—directional rank mobility (Proposition 3ii) and Shorrocks index (Proposition 4iv)—require the stronger Assumption. [b] denotes that common marginals are required unless the measure is evaluated using ranks or marginals also satisfy the usual stochastic order. Results summarize propositions in Sections 3 and 4. Proofs of the results are given in Appendix C. A 'Yes' in the 'Increasing?' column means the measure unambiguously increases (or decreases) with concordance when economies satisfy Definition 1. Applied researchers can use any measure with 'Yes' interchangeably for ordinal comparisons, subject to the restriction in the final column.

3.1. Regression

I begin by analysing two of the most widely used measures: the intergenerational earnings elasticity (IGE) and rank–rank regression slope. The IGE is the value of β, obtained from the following regression

$$\ln Y_i^K = \alpha + \beta \ln Y_i^P + \epsilon_i, \tag{3}$$

and the rank–rank regression slope, the value of ρ obtained from

$$R_i^K = \alpha + \rho R_i^P + \epsilon_i. \tag{4}$$

It is well known that the concordance order provides an ordering over covariances (Tchen 1980; Lehmann 1966) and Spearman's rank correlation. Thus, with fixed marginals, implied mobility measures (e.g., $1 - \beta$) are unambiguously smaller in more concordant economies.

PROPOSITION 1 (Linear Regression Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. If $C^A \succeq C^B$, then*

i. The intergenerational earnings elasticity (IGE) is ordered: $\beta^A \geq \beta^B$; and

ii. The rank–rank slope is ordered: $\rho^A \geq \rho^B$.

Both increase relative to the concordance ordering.

Figure 3 illustrates Proposition 1, as concordance increases (blue diamonds to orange circles), realizations cluster more tightly around co-monotonicity in both log- and rank-space. With continuous marginals, ranks are uniform and the rank–rank slope coincides with Spearman's rank correlation; hence the latter is also ordered by concordance. More generally, the intergenerational correlation, $corr(\ln Y^K, \ln Y^P)$, and the rank correlation, $corr(R^K, R^P)$, also increase with the concordance ordering.[11]

Local measures. When mobility is compared for subsets of the population, concordance need only hold on the relevant subdomain. In particular, increased concordance lowers the expected rank of children born to parents below any rank threshold and raises it for children born to parents above that threshold.

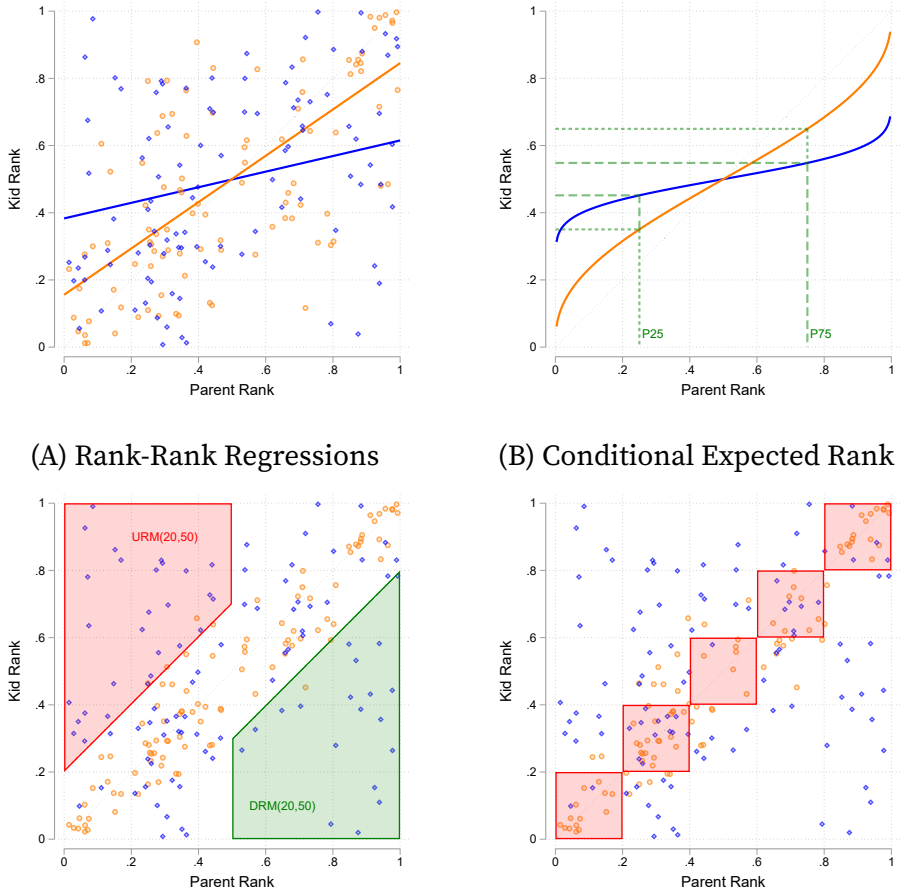
PROPOSITION 2 (Conditional Expected Rank Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. If $C^A \succeq C^B$, then for every $r \in (0, 1)$:*

i. expected child rank is lower among parents below the rank threshold, $\mathbb{E}^A\left[R^K \mid R^P \leq r\right] \leq \mathbb{E}^B\left[R^K \mid R^P \leq r\right]$, and

ii. expected child rank is higher among parents above the rank threshold, $\mathbb{E}^A\left[R^K \mid R^P > r\right] \geq \mathbb{E}^B\left[R^K \mid R^P > r\right]$.

A stronger pointwise result follows for the linear projection of the conditional expected rank. Concordance implies that the conditional expected rank curve rotates around the median.

[11] Hart (1983) propose using one minus the intergenerational correlation as a measure of mobility. Similarly, increased concordance also raises *average* non-linear persistence measures (e.g., Arellano, Blundell, and Bonhomme 2017).



(A) Rank-Rank Regressions

(B) Conditional Expected Rank

(C) Directional Mobility

(D) Shorrocks (1978) Trace Measure

FIGURE 3. Mobility Measures Under Increased Concordance

Notes: Marginal distributions are held fixed. Orange circles are drawn from an elliptical copula with a higher correlation parameter than blue diamonds, so they are more concordant. Panel (A): 100 draws and the population rank–rank regression (Proposition 1). Panel (B): population conditional expected rank (Proposition 2, Corollaries 1 & C.1); dashed green lines mark the commonly reported 25th and 75th percentiles. Panel (C): upward (red) and downward (green) rank mobility, i.e., children of below- (above-) median parents moving up (down) more than a quintile. Panel (D): Shorrocks trace regions using quintiles (red).

COROLLARY 1 (Linear Conditional Expected Rank Measures). *Let* $\widehat{\mathbb{E}}\left[R^K \mid R^P = r\right]$ *denote the fitted value from the rank–rank regression* (4). *If* $C^A \succeq C^B$, *the fitted conditional expected rank rotates around the median:*

i. it is concordance–decreasing below the median, $\widehat{\mathbb{E}}^A\left[R^K \mid R^P = r\right] \leq \widehat{\mathbb{E}}^B\left[R^K \mid R^P = r\right] \quad \forall\, r \leq 0.5$, *and*

ii. it is concordance–increasing above the median,

$$\widehat{\mathbb{E}}^A\left[R^K \mid R^P = r\right] \geq \widehat{\mathbb{E}}^B\left[R^K \mid R^P = r\right] \qquad \forall\, r \geq 0.5\,.$$

This is a widely adopted reporting practice, for example, mobility is often summarized by the expected rank of children born to parents at the 25th or 75th percentile. For the linear projection, these measures are pinned down by the rank–rank slope and therefore inherit the median-rotation result in Corollary 1. I show that this can be extended to non-parametric pointwise conditional expectations under shape restrictions in Appendix C.

3.2. Transition Probabilities

While regression measures summarize mobility globally, practitioners often focus on specific transitions such as "rags-to-riches" or "riches-to-rags" probabilities (e.g., Corak and Heisz 1999; Chetty et al. 2014a) which capture mobility for particular subgroups. Two of the results below—directional rank mobility and, later, the Shorrocks trace (Proposition 4iv)—require a stronger notion of diagonal concentration.[12]

ASSUMPTION 1 (Strong Diagonal Concentration). *Let $C^A \succeq C^B$. The change in dependence from economy B to economy A additionally satisfies:*

(i) Directional contraction. *There exists $r^\star \in (0,1)$ such that, for almost every rank r,*

$$C^A_{K|P}(r+s \mid r) \geq C^B_{K|P}(r+s \mid r), \qquad r \leq r^\star, \quad s \geq 0, \quad r+s \leq 1, \tag{5}$$

$$C^A_{K|P}(r-s \mid r) \leq C^B_{K|P}(r-s \mid r), \qquad r \geq r^\star, \quad s \geq 0, \quad r-s \geq 0, \tag{6}$$

where $C^i_{K|P}(\cdot \mid r)$ denotes the conditional distribution of child rank given parental rank r.

(ii) Diagonal interval dominance. *For every rank interval $(a, b] \subseteq [0,1]$,*

$$\Pr^A\left(a < R^K \leq b,\ a < R^P \leq b\right) \geq \Pr^B\left(a < R^K \leq b,\ a < R^P \leq b\right). \tag{7}$$

Assumption 1 requires increases in dependence to concentrate rank transitions toward the diagonal in two senses. This rules out increases in concordance generated

[12]The additional restrictions formalize the idea that stronger dependence concentrates transitions toward the diagonal, rather than merely increasing lower- and upper-orthant dependence. Both conditions are satisfied, for example, by Gaussian and FGM copula families as their dependence parameters increase. Here and below, conditional rank distributions are understood as regular conditional distributions and the inequalities hold for almost every parental rank r.

Imposing restrictions on the joint density is standard in mobility research. Chetty et al. (2017) and Berman (2022) invoke empirical plausibility arguments that implicitly consisrent with similar conditions. Shorrocks (1978) assumes away "*empirically unlikely*" transition matrices without quasi-maximal diagonals.

by offsetting changes weakening persistence. Part (i) reduces large upward movements among low-rank parents and large downward movements among high-rank parents, which is necessary to establish monotonicity of directional mobility. Part (ii) increases the probability that parent and child remain within the same rank interval which is necessary to establish monotonicity of the Shorrocks index.

PROPOSITION 3 (Rank–Based Local Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. The following results hold.*

i. Transition Probabilities: If $C^A \succeq C^B$, then measures along the positive diagonal are concordance–increasing, while those in off diagonals are decreasing:

a. Positive diagonal: $TP^A\left[R^K > \tau^K \mid R^P > \tau^P\right] \geq TP^B\left[R^K > \tau^K \mid R^P > \tau^P\right]$ and $TP^A\left[R^K \leq \tau^K \mid R^P \leq \tau^P\right] \geq TP^B\left[R^K \leq \tau^K \mid R^P \leq \tau^P\right]$ $\forall\, \tau^K, \tau^P \in [0,1]$.

b. Off-diagonal: $TP^A\left[R^K > \tau^K \mid R^P \leq \tau^P\right] \leq TP^B\left[R^K > \tau^K \mid R^P \leq \tau^P\right]$ and $TP^A\left[R^K \leq \tau^K \mid R^P > \tau^P\right] \leq TP^B\left[R^K \leq \tau^K \mid R^P > \tau^P\right]$ $\forall\, \tau^K, \tau^P \in [0,1]$.

ii. Bhattacharya and Mazumder (2011)'s Directional Rank Mobility: If Assumption 1(i) holds, then the probability that a child's rank exceeds their parent's rank by more than s, conditional on parental rank being below τ, is concordance–decreasing:

$$URM^B(s,\tau) = \Pr^B\left(R^K - R^P > s \mid R^P \leq \tau\right) \geq \Pr^A\left(R^K - R^P > s \mid R^P \leq \tau\right) = URM^A(s,\tau)$$
$$\forall\, s \in [0,1],\ \tau \in (0, r^\star],$$

Analogously, for downward rank mobility.

iii. Conversely, the concordance ordering is characterized by the lower-left transition probabilities. That is, $C^A \succeq C^B$ if and only if

$$TP^A\left[R^K \leq \tau^K \mid R^P \leq \tau^P\right] \geq TP^B\left[R^K \leq \tau^K \mid R^P \leq \tau^P\right] \quad \forall\, \tau^K \in [0,1],\ \tau^P \in (0,1]. \tag{8}$$

Proposition 3 shows that under concordance all diagonal transitions increase, all off-diagonal transitions decrease. Conversely, uniform ordering of the lower-left transition probabilities across all thresholds characterizes the concordance order. Unlike transition probabilities, directional mobility integrates over regions of the copula (see Figure 3 Panel C) and requires the stronger regularity condition that mass shifts uniformly toward the diagonal (Assumption 1).

3.3. Fixed Rank Comparisons for Subpopulations

Empirical studies often evaluate ranks against a fixed external benchmark rather than within their own population. Examples include ranking children by their position in the parent cohort's income distribution; comparing subnational mobility against the national income distribution (e.g. Chetty et al. 2014a; Bütikofer, Dalla-Zuanna, and Salvanes 2022; Deutscher and Mazumder 2020; Bell, Blundell, and Machin 2023); or comparing outcomes across racial groups (in the spirit of Bayer and Charles 2018).[13] Let

$$\widetilde{R}^K = F_{\widetilde{Y}}\left(F^{-1}_{Y^K}\left(R^K\right)\right), \tag{9}$$

be the child's rank in a fixed reference distribution $\widetilde{Y}$, e.g. their ranking in the parent's income distribution or national distribution. Since this is weakly increasing in R^K and the concordance order is preserved under componentwise increasing transformations, fixed-reference ranks preserve the concordance ordering. When the transformation is strictly increasing, the copula itself is unchanged. Consequently, all rank-based exchange mobility measures remain monotone in concordance even under fixed benchmark rankings (an analogous argument applies for parent ranks). Those with higher income ranks in their own distribution are also higher in the fixed reference distribution of incomes.

3.4. Axiomatic Measures of Exchange Mobility

A complementary approach axiomatizes mobility measures (similarly to inequality measurement). They precisely define minimal and maximal mobility as well as necessary properties of mobility measures. Axiomatic frameworks (e.g. Fields and Ok 1996; Cowell and Flachaire 2018) specify distance metrics on parent–child distributions that uniquely define a mobility index. Likewise, D'Agostino and Dardanoni (2009) derive a rank-based concordance index from ordering axioms, and Shorrocks' (1978) trace measure captures the probability of escaping one's parental "class". Under common marginals each of these indices hinges solely on the underlying dependence structure. Therefore, they are all monotone in the concordance order.

PROPOSITION 4 (Axiomatic Measures). *Economies A and B have identical marginals, but*

[13] Nonparametric estimation of these fixed-benchmark ranks corresponds to absolute mobility measures (Deutscher and Mazumder 2023).

different rank dependence denoted by copulas C^A and C^B, respectively, with $C^A \succeq C^B$. The following axiomatic measures of exchange mobility are decreasing in the concordance order:

i. *Fields and Ok (1996, 1999)'s measures of absolute differences $\mathbb{E}\left[\left|Y^K - Y^P\right|\right]$ and its decomposition.*
ii. *D'Agostino and Dardanoni (2009)'s concordance measure of rank mobility for global matrices and the extension to their preorder.*
iii. *Cowell and Flachaire (2018)'s measure of the distance between relative positions, $\mathbb{E}\left[\frac{1}{\alpha(\alpha-1)}\left(Y^K/\mathbb{E}[Y^K]\right)^{\alpha}\left(Y^P/\mathbb{E}[Y^P]\right)^{1-\alpha}\right]$ where α controls the relative weight on upwards movements compared to downwards movements.*

Additionally, if Assumption 1(ii) holds then

iv. *Shorrocks (1978)'s trace measure, $\frac{q-trace(\mathcal{Q})}{q-1}$, for stochastic matrix $\mathcal{Q}$ with elements $\mathcal{Q}_{i,j} = \Pr\left(\frac{j-1}{q} < R^K \leq \frac{j}{q} \mid \frac{i-1}{q} < R^P \leq \frac{i}{q}\right)$.*[14]

3.5. Proxies and Measurement Error

In practice, researchers often observe proxies—education, occupation, health status—rather than lifetime income. Concordance orderings are preserved when the mapping from true to observed outcomes and the distribution of measurement error are common across the economies being compared.[15]

PROPOSITION 5 (Concordance of proxies). *Let a proxy E in economy $i \in \{A, B\}$ be determined by*

$$E_i^s = H^s\left(R_i^s, \varepsilon_i^s\right), \qquad s \in \{K, P\},$$

where the functions $H^s(\cdot,\cdot)$ are common across economies. Assume each $H^s(\cdot,\cdot)$ is weakly increasing in each argument, and that $(\varepsilon_i^K, \varepsilon_i^P)$ is independent of (R_i^K, R_i^P) and has the same joint distribution in economies A and B. If $C^A \succeq C^B$, then $(E_A^K, E_A^P) \succeq_c (E_B^K, E_B^P)$.

Moreover, the proxy marginals are identical across economies. Hence, under the checkerboard convention for mass points, the corresponding proxy copulas satisfy $\widetilde{C}_E^A \succeq \widetilde{C}_E^B$.

This result includes discrete proxies, such as categorical education groups or occupation groupings, and accommodates a variety of empirically relevant proxy

[14] This is obtained by discretizing ranks into q groups. For simplicity, I consider the case where q groups are of equal size, but this is not integral to the result.

[15] Writing the proxy as a function of ranks rather than income levels imposes no additional restriction when the income marginals and proxy mapping are held fixed across economies: for continuous marginals, $Y^s = (F_Y^s)^{-1}(R^s)$, so any $E^s = h^s(Y^s, \varepsilon^s)$ can be rewritten as $E^s = H^s(R^s, \varepsilon^s)$.

frameworks. In particular, it captures cases where education proxies income and where latent ability drives both education and income.

Example 1: Human capital determines incomes. Suppose increased human capital directly increases lifetime incomes (Mincer 1974) where $Y = H(E) + \varepsilon$. Inverting yields $E = H^{-1}(Y - \varepsilon)$. Treating ε as a vector allows for a general functional form for H and Y which can accommodate cases where the variance of the income residual increases in education as is standard in empirical work.

Example 2: Ability determines education and income. Alternatively, let a latent skill S determine both education and income: $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

In each example the function H can differ across generations. A related latent-rank interpretation is used by Asher, Novosad, and Rafkin (2024), who treat observed education as a coarse measure of an underlying rank distribution. Parent-child joint distributions, e.g. $F(Y^K, E^P)$ or $F(E^K, E^P)$, inherit the dependence structure of the copula $C(R^K, R^P)$, so concordance results extend immediately to proxies.

3.5.1. (Non-Classical) Measurement Error

Although measurement error can alter measured dependence, concordance orderings are preserved when the measurement structure is common across economies: mismeasured incomes then act as proxies in the sense of Proposition 5. Thus, even though mobility measures such as rank–rank slopes or elasticities may be biased in level, their concordance-implied ordering across economies is preserved under a common measurement structure.

COROLLARY 2 (Concordance under mismeasurement). *Let observed income in economy $i \in \{A, B\}$ satisfy*

$$\widetilde{Y}_i^s = G_s\left(R_i^s, \varepsilon_i^s\right), \qquad s \in \{K, P\},$$

where the functions $G_s(\cdot, \cdot)$ are common across economies and weakly increasing in each argument. Assume

$$(\varepsilon_i^K, \varepsilon_i^P) \perp (R_i^K, R_i^P),$$

and that the joint distribution of $(\varepsilon_i^K, \varepsilon_i^P)$ is the same in economies A and B. Then, if $C^A \succeq C^B$, the copulas of the corresponding observed income pairs satisfy $\widetilde{C}^A \succeq \widetilde{C}^B$.

When measurement quality differs across economies, the ordering need not be preserved. Importantly, assuming observed measures are monotone functions of true earnings does not rule out non-classical measurement error (in the spirit of Bound et al. 1994) and standard noise structures satisfy Corollary 2 directly. Classical additive error in income levels, $\widetilde{Y}_i^s = Y_i^s + \varepsilon_i^s$, and error-in-ranks specifications of the form $\widetilde{R}_i^s = F_{\widetilde{s}}\left(F_s^{-1}(R_i^s) + \varepsilon_i^s\right)$ (Chetverikov and Wilhelm 2023) both yield observed measures that are weakly increasing in true lifetime income, satisfying the monotonicity requirement with generation-specific error distributions held fixed across economies.

Life-cycle biases. Systematic variation in the link between current and lifetime earnings (Jenkins 1987; Nybom and Stuhler 2016) produces:

$$\widetilde{Y}_i^K = G_K\left(F_K^{-1}(R_i^K) + \mu_{\varepsilon^K} + \sigma_{\varepsilon^K}\varepsilon_i^K\right), \qquad \widetilde{Y}_i^P = G_P\left(F_P^{-1}(R_i^P) + \mu_{\varepsilon^P} + \sigma_{\varepsilon^P}\varepsilon_i^P\right),$$

which also satisfies the Corollary's restrictions. Concordance orderings are preserved under these life-cycle biases when the corresponding measurement equations and noise distributions are common across the economies being compared. Similar points are made by Bhattacharya and Mazumder (2011) and Kitagawa, Nybom, and Stuhler (2018), who derive results for generalised life-cycle bias measurement equations, in the context of transition probabilities and rank–rank correlations, respectively.

4. Decomposing the Roles of Dependence and Marginal Distributions

Previous results assumed common marginals to isolate variation in the dependence structure alone. This corresponds to comparing hypothetical variants of a single economy. Comparisons across space and time—such as asking whether the U.S. is more mobile today than in 1965 (Chetty et al. 2017) or whether the U.S. is more mobile than Canada (Corak 2020)—typically involve different marginal income distributions. I now show that any change in a mobility measure can be decomposed into a component due to changing marginals and a component due to changing dependence.

Decomposition. By Sklar's theorem, any joint distribution can be written $F(y^K, y^P) = C\left(F_K(y^K), F_P(y^P)\right)$, where C is the copula and F_K, F_P are the marginal distributions. A mobility measure M therefore depends on both components: $M = M(C, F_K, F_P)$. For two

economies A and B, the change in any measure can be decomposed as:

$$\begin{aligned}\Delta M = M^B - M^A &= M(C^B, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A) \\ &= \underbrace{\left[M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)\right]}_{\text{Copula component (varying dependence)}} + \underbrace{\left[M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)\right]}_{\text{Marginal component (varying marginals)}} . \quad (10)\end{aligned}$$

The first term captures the change in mobility due to differences in the dependence structure, holding marginals fixed at economy B's distributions. The second term captures the change due to differences in marginals, holding the copula fixed at economy A's dependence structure.[16] The choice of the reference marginal or copula is application dependent like the choice of reference period in price indices.[17]

This decomposition clarifies when a concordance ordering applies. If $C^B \succ C^A$, the copula component of each measure in Table 1 has the sign implied by the concordance order. For the marginal component, which holds the copula fixed, all rank–based measures are invariant—since continuous ranks have uniform marginals by construction, changes in income distributions leave the rank–rank correlation, transition matrices, and Shorrocks index unchanged. Level–based measures, however, respond to marginal shifts even when dependence is unchanged. Appendix D illustrates the decomposition in closed form using the bivariate log-normal, showing that the rank–rank correlation depends only on the correlation parameter ρ while the IGE depends on $\rho \cdot \sigma_K/\sigma_P$, so rising inequality can reverse the IGE's trend even as rank-dependence weakens.

Implications. Rank–based measures isolate positional dependence and are ordered whenever copulas are ordered, regardless of differences in income distributions, as

[16] For distributions with mass points, throughout this decomposition the copula refers to the checkerboard extension defined in Section 2 and the marginal distributions remain the observed marginals.

[17] An alternative decomposition holds marginals at A and the copula at B, averaging yields:

$$\begin{aligned}\Delta M = &\underbrace{\left[\tfrac{1}{2}\left(M(C^B, F_K^B, F_P^B) - M(C^A, F_K^B, F_P^B)\right) + \tfrac{1}{2}\left(M(C^B, F_K^A, F_P^A) - M(C^A, F_K^A, F_P^A)\right)\right]}_{\text{Copula component}} \\ &+ \underbrace{\left[\tfrac{1}{2}\left(M(C^A, F_K^B, F_P^B) - M(C^A, F_K^A, F_P^A)\right) + \tfrac{1}{2}\left(M(C^B, F_K^B, F_P^B) - M(C^B, F_K^A, F_P^A)\right)\right]}_{\text{Marginal component}} . \quad (11)\end{aligned}$$

are proxy-based indices mapping ranks to observed categories.[18] They are the natural choice when the object of interest is positional mobility across economies. Level–based measures (e.g., the IGE or IGC) combine both sources of variation, so the marginal component can offset or reverse a concordance-ordered copula component.[19]

5. Interpreting Concordance In a Model of Human Capital Investment

Concordance is a sufficient statistic for the positional mobility component of standard mobility measures. This section asks what is sufficient for concordance itself. In a Becker and Tomes (1979)–Loury (1981) model of human capital investment, the answer is the *incidence* of investment or skill gains across economies. Skill gains that rise with parental resources lower mobility by every measure ordered by concordance, while incidence that falls raises it, and non-monotone incidence produces crossing copulas under which measures can disagree.

Setup. A family consists of a parent and child indexed by P and K. In economy $s \in \{A, B\}$, parents with resources Y^P receive a transfer $T^s(Y^P)$ and choose investment, I, in their child's human capital to solve

$$\max_I \; U\left(Y^P + T^s(Y^P) - I\right) + \delta^s \, \mathbb{E}_\theta\Big[U\Big(W^s\big(\theta f^s(I,\, Y^P)\big)\Big)\Big], \tag{12}$$

where f^s is the skill-formation technology (positive and weakly increasing), W^s is a strictly increasing wage schedule mapping human capital to income, δ^s is the altruism parameter.[20] Luck, denoted by θ, is realised after investment, is independent of parental resources and family choices, and has the same distribution in both economies.[21]

[18] This implies rank–based alternative versions of measures are ordered. For example, the Fields and Ok (1996) measure computed using ranks corresponds to the Bartholomew (1973) average jump measure. The assumption that proxies depend on ranks, not levels, can be tested in auxiliary datasets without information on two generations.

[19] Under stochastic dominance of the marginal distributions, a general ranking for all supermodular measures obtains (see Meyer and Strulovici 2013). When marginals are not identical but are sufficiently close, continuity results apply analogously to Proposition 7.

[20] I abstract from parental beliefs, dynamic complementarity, multidimensional skills, and borrowing constraints (see e.g., Cunha and Heckman 2007; Caucutt and Lochner 2020; Attanasio, Meghir, and Nix 2020; Caucutt et al. 2020; Moschini 2023). Appendix L extends the main result to a multi-period skill technology with several inputs.

[21] The timing follows Loury (1981). Allowing W^s to differ across economies lets wages adapt to the supply and composition of educated workers without fully specifying general equilibrium forces (following Heckman, Lochner, and Taber 1998; Lee and Wolpin 2006). Parental human capital enters through Y^P, a small abuse of notation.

The skill index, $S^s(y) \equiv \ln f^s\big(I^{\star s}(y), y\big)$ where $I^{\star s}$ is optimal investment, is the systematic component of human capital for a child born to parents with resources y—it summarises every channel through which parental resources shape child outcomes, including the investment response to prices, transfers, and preferences.

As the wage schedule is strictly increasing, a child's rank in the income distribution coincides with their rank in the human capital distribution.[22] Log human capital is

$$\ln H^{K,s} = S^s(Y^P) + \ln \theta. \tag{13}$$

This structure makes the source of positional dependence transparent. For a child from a poor family to overtake a child from a rich family, their luck must exceed the gap between the two families' skill indices.

The copula therefore depends on how those gaps change across economies, and the natural primitive is the change in the skill index,

$$\Delta(y) = S^A(y) - S^B(y), \tag{14}$$

which I refer to as the investment *incidence* of a change in the economic environment.

5.1. When Does the Concordance Order Arise?

PROPOSITION 6 (Incidence and Concordance). *Suppose parental resources have a common continuous distribution with compact support,* $\ln \theta$ *has an everywhere positive, bounded, and continuous density, and* S^A *and* S^B *are bounded.*

i. If Δ *is nondecreasing in parental resources, then* $C^A \succeq C^B$*. If* Δ *is nonincreasing, then* $C^B \succeq C^A$*.*

ii. Average incidence is irrelevant to positional dependence: C^A *depends on* Δ *only through its deviations from the mean,* $\Delta - \mathbb{E}[\Delta(Y^P)]$*. In particular, if* Δ *is constant, then* $C^A = C^B$*.*

iii. The wage schedule affects positional dependence only through investment: given f^s *and* $I^{\star s}$*, the copula* C^s *is the same for every strictly increasing* W^s*.*

Intuitively, who gains skills matters. When children from higher-income families gain at least as much in log skill as children from lower-income families, a larger level of luck is required for their positions to reverse, and concordance rises. When poorer families gain proportionally more, skill gaps narrow and concordance falls.

[22] As Lochner and Park (2024) emphasise, intergenerational income mobility will differ from skill mobility if wages are stochastic given human capital.

Proportional gains in skill leave every log skill gap unchanged, and so also the copula, while changes in the skil price stretch the income distribution without reordering children, so any effect of wages on positional dependence must operate through the response of investment. The proof is in Appendix L.

Effects on the level of investment or skill are therefore insufficient for positional mobility despite being the relevant statistic for growth and absolute mobility. What matters is not whether a change to the economy raises children's outcomes, but whether it raises them proportionally more for children of low– or high–income parents. Estimates of how treatment effects vary with family background (e.g., Jackson, Johnson, and Persico 2016; Chetty and Hendren 2018a; Biasi 2023) are the natural empirical counterpart. In this setting, estimating a policy's effect on adult outcomes across the full parental income distribution, as in Havnes and Mogstad (2015), characterizes positional mobility.

To make the incidence condition concrete, consider log utility, Cobb–Douglas technology $f^s(I, y) = I^{\alpha^s} y^{\gamma^s}$, and isoelastic wages $W^s(H) = \omega^s H^{\kappa^s}$. Here α^s measures the productivity of investment, γ^s the direct effect of parental resources, and κ^s the return to skill. Optimal investment is a constant share of disposable resources, $I^{\star s}(y) = c^s[y + T^s(y)]$, with $c^s = \delta^s\kappa^s\alpha^s/(1 + \delta^s\kappa^s\alpha^s)$.[23]

COROLLARY 3 (Investment, Incidence, and Concordance). *In the log-linear model:*

i. Skill-formation technology. *If* $T^A = T^B = 0$, $\alpha^A \geq \alpha^B$, *and* $\gamma^A \geq \gamma^B$, *then* $C^A \succeq C^B$.

ii. Patience, proportional transfers, and the return to skill. *If* α^s *and* γ^s *are common,* $T^B = 0$, *and* $T^A(y) = \tau y$ *with* $\tau \geq 0$, *then* $C^A = C^B$ *for any* δ^s *and* κ^s, *although investment is higher at every parental income level whenever* $\delta^A\kappa^A \geq \delta^B\kappa^B$. *The IGE satisfies* $\beta^A = (\kappa^A/\kappa^B)\,\beta^B$.

iii. Equal cash transfers. *If all other primitives are common,* $T^B = 0$, *and* $T^A(y) = t > 0$, *then* $I^{\star A}(y) > I^{\star B}(y)$ *for all* y, *yet* $C^B \succeq C^A$.

Corollary 3 makes the insufficiency of investment effects concrete. Each mechanism in parts (ii) and (iii) raises investment at every parental income level, yet part (ii) leaves concordance unchanged and part (iii) lowers it. Concordance rises only in part (i), because a more productive technology steepens the gradient of log skill in log parental resources—$\Delta(y)$ rises with $\ln y$.

[23] This gives rise to the standard log-linear structural specification for the IGE.

5.2. When Does the Concordance Order Fail?

Not all economically relevant changes have monotone incidence, however. Consider a program targeted at disadvantaged families that coincides with rising returns to elite schooling or increased neighbourhood sorting. Children at both ends of the parental distribution gain, while children in the middle gain least. Incidence is U-shaped and Proposition 6 delivers no order. In fact, for changes of this kind the copulas must cross.

COROLLARY 4 (Non-Monotone Incidence and Crossing Copulas). *Suppose Δ is continuous and children of both the lowest- and highest-resource parents gain more than the average child,*

$$\min\left\{\Delta(\underline{y}),\ \Delta(\overline{y})\right\} > \mathbb{E}\left[\Delta(Y^P)\right].$$

Then, for all sufficiently small $z > 0$, the economy with skill index $S^B + z\Delta$ is not ordered by concordance relative to B.

The condition here implies a non-monotonicity in relative incidence in both tails of the parental income distribution. Thus, persistence at the bottom falls while persistence at the top rises—consistent with the disagreement between bottom and top persistence across commuting zones documented in the introduction. The same logic disciplines the interpretation of policy. School finance equalisation that reallocates resources toward poorer districts (Jackson, Johnson, and Persico 2016; Biasi 2023) lowers concordance unambiguously when its proportional effects decline with parental resources, and generates crossing copulas only when those effects are non-monotone. Distinguishing the two turns on how effects vary across the full parental distribution, rather than on average effects. The proof is in Appendix L, which shows this result holds under a weaker, but less transparent, sufficient condition.

6. When Concordance Fails

Sections 2 and 3 establish that concordance provides a unifying framework. However, concordance is only a *partial* order—not all pairs of economies can be compared. This section examines when concordance fails and why different mobility measures then disagree, and why uniform agreement across threshold transition probabilities becomes impossible. Section 7 addresses how researchers should proceed in such cases.

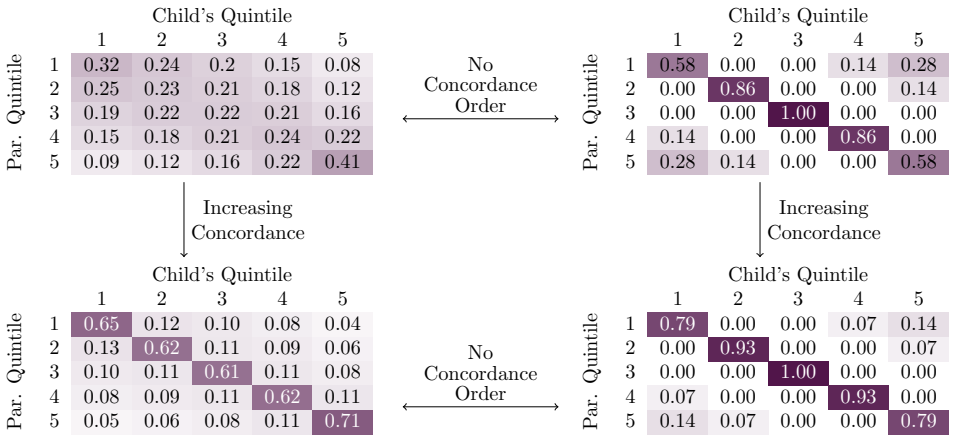


FIGURE 4. An Example of Concordance as a Partial Order

Notes: The first row is adapted from an example in Berman (2022). Each row produces approximately equal values for the rank correlation.

6.1. The Limits of Concordance

When can concordance fail to rank two economies? Figure 4 illustrates this directly. Comparing vertically, the pairs of transition matrices increase in the concordance order. However, no concordance ordering exists *between* horizontal comparisons. The matrices in each row produce approximately equal rank correlations (0.32 and 0.65, respectively), yet differ substantially in their transition probabilities or mass along the trace. In these cases a single global measure does not pin down all mobility measures.[24]

6.2. Why Mobility Measures Disagree

Each measure weights different regions of the joint distribution and any mobility measure can be expressed as a weighted integral of the joint density

$$M_M = \int \int \phi_M \left(Y^K, Y^P\right) f \left(Y^K, Y^P\right) d\, Y^K d\, Y^P, \tag{15}$$

where $\phi_M \left(Y^K, Y^P\right)$ is a mobility-measure specific intensity assigning importance to parent–child income pairs. When comparing economies A and B with different copulas

[24] Berman (2022) argues copulas in the left column are empirically plausible (comparing to estimates in Jäntti et al. 2006 and Chetty et al. 2017), but rejects the right-column "all-or-nothing" dynamics. This illustrates that applied work already uses substantive restrictions on plausible patterns of intergenerational dependence. Concordance provides a comparative ordering of such dependence structures the validity across economies is an empirical question.

but common marginals, the difference in any twice-differentiable measure can be decomposed as:

$$M_M^A - M_M^B = \int_0^1 \int_0^1 \underbrace{\frac{\phi_{M,12}\left(F_K^{-1}(R^K), F_P^{-1}(R^P)\right)}{f_K\left(F_K^{-1}(R^K)\right)\, f_P\left(F_P^{-1}(R^P)\right)}}_{W_M(R^K,R^P)} \left[C^A(R^K, R^P) - C^B(R^K, R^P)\right] dR^K\, dR^P. \tag{16}$$

When concordance orders economies, $C^A(R^K, R^P) - C^B(R^K, R^P) \geq 0$ everywhere in equation (16), so all measures agree despite different weights W_M. When concordance fails, the difference in copulas changes sign across regions. In this case, one measure may place high weight on regions where $C^A > C^B$ while another emphasises regions where $C^A < C^B$, producing conflicting rankings. Measure choice then embeds an implicit judgment over what constitutes desirable or relevant mobility (Fields and Ok 1996; Jäntti and Jenkins 2015; Ray and Genicot 2023).

This explains the different levels of sensitivity of measures to concordance violations. Some measures average dependence broadly across the joint distribution. The rank–rank slope, for instance, has an intensity that assigns uniform weight $W_\rho = 12$ everywhere—it is, in effect, an unweighted average of ΔC over the unit square. Such measures are relatively insensitive to local concordance violations because positive and negative regions can offset. Other measures zoom in on specific regions. Transition probabilities at particular thresholds have indicator-function intensities—they depend on ΔC at a single point, with zero weight elsewhere. These measures are maximally sensitive to which side of a sign change they happen to evaluate. The intergenerational elasticity falls between these extremes: its intensity loads most heavily on the tails (families furthest from the mean, $\phi_M\left(Y^K, Y^P\right) = [\ln Y^K - \overline{\ln Y^K}][\ln Y^P - \overline{\ln Y^P}]$), making it sensitive to concordance violations concentrated in those regions.

6.3. Disagreement is Generic

Inspecting equation (16) shows that it is always feasible to construct two measures that disagree when the copulas are not concordance ordered. But concordance provides a much sharper characterisation of disagreement among *widely used* measures.

First, for threshold transition probabilities, concordance is not just sufficient for uniform agreement—it is necessary. Proposition 3 shows that lower-left transition probabilities order the economies consistently at all admissible cutpoints if and only if

the copulas are concordance ordered. Second, concordance violations can manifest in ranking reversals even for pre-specified discretisations including standard empirical objects (e.g., quintile, decile, or ventile transition matrices) which I formalize in Appendix E.1. Whether concordance holds between empirical copulas can be assessed directly by testing the pointwise copula inequalities in the spirit of moment inequality methods; Section 8 implements this procedure and Appendix J provides details.

7. What To Do When Concordance Fails

Section 3 establishes that concordance is necessary for uniform agreement of transition probabilities. This section provides practical guidance for researchers. How sensitive are orderings to small violations (7.1)? When they are substantive, what should be reported (7.2)? Finally, when a single ranking is needed, how should the partial order be completed (7.3)?

7.1. Approximate Concordance

Concordance results are not a knife-edge property. If two copulas nearly satisfy concordance, with small violations, measures are approximately ordered.

PROPOSITION 7 (Continuity and Approximate Concordance). *Let M be any real-valued function of the copula C (with fixed marginals) such that*

i. (Concordance Monotonicity) for any copulas C', C'', if $C' \succeq C''$ then $M(C') \leq M(C'')$

ii. (Uniform Continuity) M is uniformly continuous with respect to the copula sup-norm $||C' - C''||_\infty \equiv \sup_{(u,v)\in[0,1]^2} |C'(u,v) - C''(u,v)|$.

If there exists a copula $C^\dagger$ such that $C^A \succeq C^\dagger$ and $||C^B - C^\dagger||_\infty \leq \epsilon$ (i.e., C^B is obtained as an ϵ-perturbation of a copula $C^\dagger$ that is dominated by C^A in the concordance order), then

$$M(C^A) \leq M(C^B) + \omega_M(\epsilon)\,, \tag{17}$$

where $\omega_M(\epsilon) \equiv \sup\{|M(C') - M(C'')| : ||C' - C''||_\infty \leq \epsilon\}$ is the measure M's modulus of continuity. In particular, if M is Lipschitz continuous under the copula sup-norm ($||\cdot||_\infty$) with constant L_M, then

$$M(C^A) \leq M(C^B) + L_M\epsilon\,. \tag{18}$$

The constant L_M governs each measure's sensitivity to violations and Appendix F derives it for the measures in Table 1, highlighting that measures concentrated on a

small region are often more sensitive. This helps explain the tighter co-movement of rank–based measures in empirical work (Deutscher and Mazumder 2023).

7.2. Reporting When Concordance Fails

When concordance fails substantively—copulas exhibit non-trivial offsetting changes across regions—reporting only a single global measure is inadequate. As the empirical application below shows, researchers using different discretisations can reach opposite conclusions about the same data.

This problem has an intuitive solution: report familiar estimands—rank–rank slopes or transition probabilities—computed locally in the regions where concordance holds and fails. This allows each reader to evaluate whether the concordance violation occurs in a region that matters for their preferred measure.[25]

The concordance test itself provides the necessary guidance on *where* to report. The moment inequality test in Section 8 evaluates ΔC on a grid of rank-space. The grid points where violations are detected—where $\hat{C}^A(u,v) < \hat{C}^B(u,v)$—identify the regions where dependence shifted in the direction opposite to the global trend. These regions define a natural partition for local reporting.

A reporting protocol.

(a) *Test concordance*. Apply a test of concordance, for example the moment inequality test in Section 8).

(b) *If concordance holds*: report any preferred global measure and note that all concordance-monotone measures agree on the ranking. A single statistic suffices.

(c) *If concordance fails*:

  (i.) Identify the regions of rank-space where ΔC changes sign.

  (ii.) Report *local rank–rank slopes* within each region. Local slopes do not privilege one reader's preferred transition over another's—making them neutral in the sense of uniform weighting in rank space.[26]

  (iii.) Supplement with *transition probabilities* at standard thresholds (e.g., quintile boundaries) for readers who focus on specific mobility pathways. This also

[25] This reporting strategy can be formalised as a communication–admissibility requirement in the sense of Andrews and Shapiro (2021) for users with different preferred measures (Appendix G).

[26] Formally, the rank–rank slope's intensity kernel is constant ($W_\rho = 12$ everywhere), so its local version provides an unweighted average of ΔC within each region. For any reader whose preferred measure has a smooth intensity, local slopes approximate local contributions up to an error shrinking in granularity (see Appendix F).

provides information for readers whose preferred measures have concentrated rather than smooth intensities.

(d) *Optionally*: report the Lipschitz bounds from Appendix F so readers can assess the maximum effect of the concordance violation on their preferred measure.

The partition into regions need not be fine. For broadl -averaging measures (rank–rank slopes, intergenerational correlations), a coarse, data–informed, partition captures most of the relevant heterogeneity. For concentrated measures (specific transition probabilities), the point evaluations in step (c) provide exact information.

7.3. Completing the Concordance Order

When concordance definitively fails and a single ranking is nevertheless required—for example, to construct a time series or country ranking—researchers need a principled tie-breaking rule. I discuss two approaches.

Restricting the comparison set. The first approach limits attention to economies that can be ranked by concordance. Many parametric copula families naturally induce concordance orderings as dependence parameters vary (Joe 1997, ch 4). This directly rules out shifts like those in Figure 4 and retains only comparisons for which all concordance-monotone measures agree. The cost is loss of generality.

Axiomatic completion. The second approach constructs an explicit tie-breaking rule. D'Agostino and Dardanoni (2009) provide axiomatic foundations for completions in the context of discrete transition matrices, prioritising rank distance: mobility increases with the magnitude of rank changes, regardless of location. This yields the following minimum-distance completion.

DEFINITION 3 (A Minimum Distance Completion of the Concordance Order). *For any empirical copula $C(u, v)$, construct the following summary statistic by mixing between the independent copula, $C^{\perp}_{uv} = uv$, and the Frechet upper bound, $C^{+}(u, v) = \min(u, v)$:*

$$M_{\lambda} = \lambda^{\star} \in argmin_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C(u, v) - ((1 - \lambda)\, C^{+}(u, v) + \lambda\, C^{\perp}_{uv}) \right]^2 du\, dv. \tag{19}$$

Intuitively, this asks what mixture of perfect immobility (Hoeffding or Frechet bound C^{+}) and independence ($C^{\perp}$) best approximates observed mobility? The weight on independence, $\lambda^{\star}$, provides the mobility index—higher values indicate more mobility.

The completion reduces to pure concordance comparisons when those exist and is used in related contexts. Fernández and Rogerson (2001) and Abbott et al. (2019) use an identical minimum-distance estimator to measure marital sorting. Chiappori et al. (2025) use an odds-ratio index axiomatised for assortative mating which is equivalent to concordance in discrete tables, providing alternative normative foundations. An alternative welfare-consistent completion, in which a planner with submodular preferences over parent–child incomes ranks economies, is discussed in Appendix H.

Connection to inequality measurement. Concordance is the analogue to Lorenz dominance for mobility.[27] The parallel extends to completions. Just as intersecting Lorenz curves require specifying a social welfare function to produce a complete ranking (Atkinson 1970), crossing copulas require analogous normative choices—or, alternatively, the local reporting protocol described above.

8. Revisiting Intergenerational Mobility Over the 20th Century

Having established theoretical conditions under which mobility measures agree, I now test these predictions using U.S. data across the 20th century.

To link the theoretical results explored above to the study of intergenerational mobility, I construct cohort-specific mobility estimates for U.S. males born between 1910 and 1980. To limit life-cycle bias I restrict the sample to fathers and sons aged between 30 and 50. Following Jácome, Kuziemko, and Naidu (2025), I pool all available surveys that report respondents' current family income together with race, father's occupation, and region of birth or childhood and impute parental income from auxiliary data sources (primarily the Census) using race, occupation, and region.[28] This yields a repeated cross-section that is nationally representative and consistent over time which allows me to document trends in intergenerational mobility measures over the 20th

[27] A similar connection was first proposed by Dardanoni (1993). Atkinson and Bourguignon (1982) highlight concordance-increasing exchanges in the context of multi-dimensional inequality: when social welfare is utilitarian and sub-modular at the level of parent–child incomes, increased concordance lowers both mobility and social welfare.

[28] For linear estimating equations this is a Two-Sample Instrumental Variable design. Like many studies of mobility in a historical context (e.g., Collins and Wanamaker 2022; Ward 2023), this application uses both self-reported incomes and a proxy approach to construct linked parent–child outcomes. Corollary 2 shows that comparisons are robust to proxies or measurement error that are common across the economies being compared. Because the auxiliary data used to impute parental income differ across cohorts, this holds only approximately here, and differences between cohorts may partly reflect changes in how well the proxy predicts income.

century (see also, Davis and Mazumder 2024).

8.1. Concordance Mobility

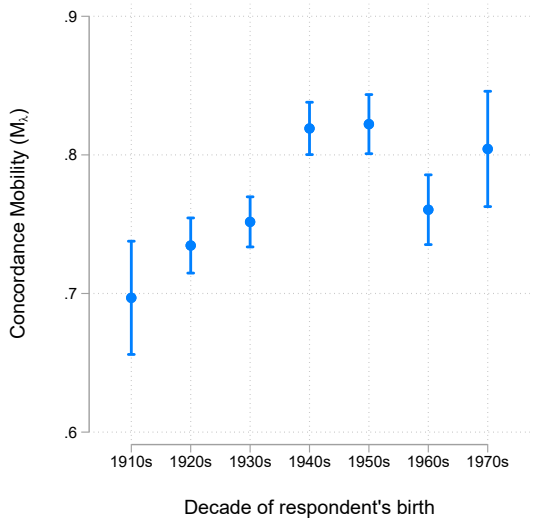


FIGURE 5. Concordance Mobility Over the 20th Century

Notes: Author's estimates from 15 combined US data sources (see text, Jácome, Kuziemko, and Naidu 2025 and Appendix I for details) by birth decade for respondents ages 30–50. Parental income predicted using family income conditional on father's occupation, race, and region (South vs. elsewhere) from auxiliary data (often the census) as close as possible to the respondents' tenth birthday (see Jácome, Kuziemko, and Naidu 2025 sec. III.B for more details). Sample weights are used and I reweight each birth cohort (i.e., decade) so that it has representative race shares. Estimate of concordance mobility measure in (19) and 95% confidence interval from 400 bootstrap replications.

Before turning to the formal tests, I summarise mobility across cohorts using the cardinal concordance measure from the completion in equation (19). Figure 5 plots this measure over the 20th century. Mobility rises from 0.70 for the 1910s cohort to 0.82 for the 1940s and 1950s cohorts, then declines for the 1960s cohort before recovering to 0.80 for the 1970s cohort.[29] The cardinal measure summarises the copula in a single number and with a weighting reflecting average rank changes. Section 8.2 tests which of these movements are robust to that weighting.

[29]The decline for the 1960s cohort is consistent with Davis and Mazumder (2024), who document falling intergenerational mobility for cohorts born in the late 1950s and early 1960s relative to those born a decade earlier. Section 8 shows that the copulas for these cohorts cross. Section 5 shows that such a decline requires non-monotone incidence, such as skill gains that rise with parental resources across most of the distribution but stall at the very top. Identifying the specific mechanisms is beyond this paper's scope.

8.2. Testing the Concordance Order

To test the order, I use the fact that $C^A \succeq C^B$ requires $C^A(u,v) \geq C^B(u,v)$ at every point in the unit square—that is, cohort A places at least as much mass as cohort B in every lower-left region of rank-space. Because outcomes contain mass points, I estimate each copula with the empirical checkerboard copula (Section 2). I evaluate $C^A - C^B$ on a 20×20 grid, studentise each inequality, and take the minimum as the test statistic, with critical values from generalized moment selection (Andrews and Soares 2010). Testing both directions, I classify a pair as consistent with an ordering if exactly one null is rejected, unordered if both are, and indistinguishable if neither is. Since any violation on the grid violates concordance, only rejections are conclusive.[30]

TABLE 2. Testing the Concordance Order Between Birth Cohorts

	p-value, H_0:				p-value, H_0:		
Cohorts	$C^A \succeq C^B$	$C^B \succeq C^A$	Outcome	Cohorts	$C^A \succeq C^B$	$C^B \succeq C^A$	Outcome
A. Relative to 1910s				*B. Sequential decades*			
1910s–20s	0.865	0.019**	↑ mobility				
1910s–30s	0.846	0.017**	↑ mobility	1920s–30s	0.469	0.178	Indistinguishable
1910s–40s	1.000	0.001***	↑ mobility	1930s–40s	0.813	0.001***	↑ mobility
1910s–50s	0.988	0.001***	↑ mobility	1940s–50s	0.001***	0.262	↓ mobility
1910s–60s	0.711	0.001***	↑ mobility	1950s–60s	0.001***	0.001***	Crossing
1910s–70s	0.860	0.005***	↑ mobility	1960s–70s	0.581	0.026**	↑ mobility

Notes: Moment-inequality tests of the concordance order following Andrews and Soares (2010), evaluated on a 20×20 grid of rank space. A denotes the earlier cohort and B the later cohort; $H_0 : C^A \succeq C^B$ tests whether the earlier cohort's copula dominates pointwise. Outcomes classify each pair at the 5% level: ↑ (↓) mobility when only $H_0 : C^B \succeq C^A$ ($H_0 : C^A \succeq C^B$) is rejected, crossing when both are rejected, and indistinguishable when neither is. Failure to reject a null does not establish the corresponding ordering. Bootstrap p-values from 999 replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2 reports the results. Panel A tests long-run comparisons against the 1910s baseline. For every subsequent decade, I reject $C^B \succeq C^A$ at the 5% level while failing to reject $C^A \succeq C^B$. Thus, the data are consistent with stronger concordance for the 1910s cohort and reject concordance in the opposite direction. Long-run increases in mobility over the 20th century are consistent with uniform increases across the joint distribution, and are robust to choices among rank–based measures.

Panel B examines sequential decades, revealing a richer picture. Three adjacent pairs are consistent with concordance. The 1930s–1940s and 1960s–1970s, show

[30] This is standard in tests of stochastic dominance.

declining concordance (rising mobility), while the 1940s–1950s show the reverse. The remaining two pairs are unordered, but for different reasons. For the 1950s–1960s, I reject concordance in both directions as the copulas cross, so dependence strengthened in some regions of the joint distribution while weakening in others. For the 1920s–1930s, I reject neither direction, so the two cohorts are statistically indistinguishable at this grid resolution. Theory predicts that mobility measures may disagree where copulas cross (here, only the 1950s–1960s), while agreeing wherever concordance orders the pair. I return to this prediction below.

The concordance test disciplines the interpretation of Figure 5. Where consistent with an ordering, M_λ moves in the implied direction in eight of nine cases. For the 1940s–1950s, the rise is just 0.003, and the ordering is not ruled out, but neither is it confirmed. When copulas cross (1950s–1960s), M_λ is a useful but imperfect summary. This matters because its fall between these cohorts is the largest in the series, yet mobility measures need not agree on its direction of change.

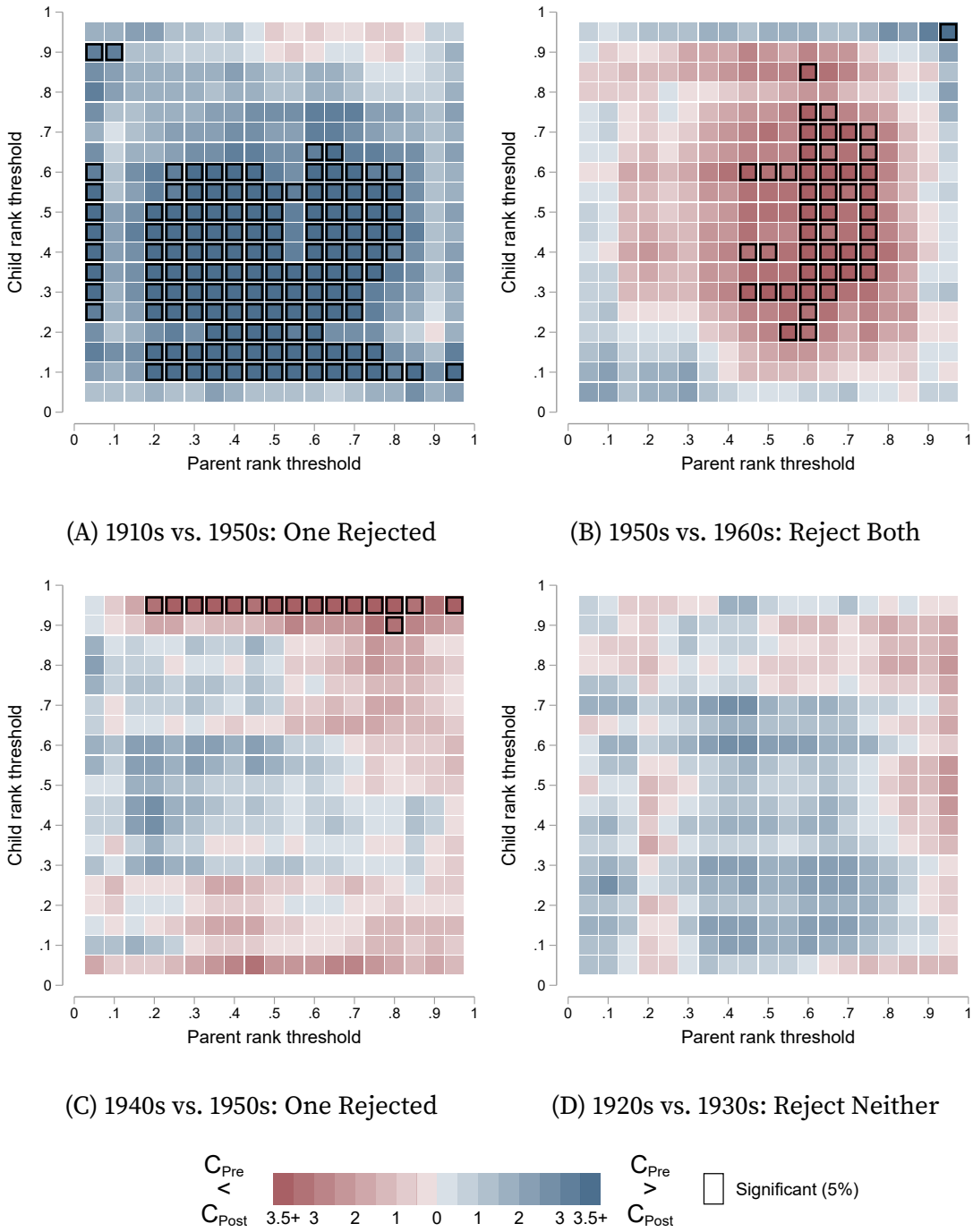
8.3. Where Do Cohorts Differ? Locating Concordance Violations

In order to understand which regions of the joint distribution are responsible for rejections and where mobility changes, Figure 6 shows the underlying pointwise inequalities and reports their studentized values and joint significance. Each cell on this figure is a threshold transition-probability comparison, and for a given coarseness, it captures the full family of transition probabilities.

Panel (A) reports the results for an ordered pair, the 1910s and 1950s birth cohorts. In this case, almost every cell is shaded blue indicating higher pointwise mobility in the later cohort and the only red shaded cells are not statistically significant. This shows that mobility is broad-based across parental ranks and for the majority of positional transitions it is statistically significant. This provides a useful benchmark for comparing to violations of the order.

Panel (B), in contrast, considers a case where the order is rejected in both directions. The majority of the pointwise inequalities are consistent with less mobility in the 1960s cohort and this pattern is statistically significant for parents between the median and 75th percentile of incomes. However, the significant region in the top ventile shows persistence at the very top declined.[31] The evaluation of mobility with a single statistic is therefore sensitive to the weights placed on these regions, for instance, the largest fall in

[31] One may worry that this is driven by top-coding of incomes, however, the ventile cutoffs are substantially below the top-coding threshold.



(A) 1910s vs. 1950s: One Rejected

(B) 1950s vs. 1960s: Reject Both

(C) 1940s vs. 1950s: One Rejected

(D) 1920s vs. 1930s: Reject Neither

FIGURE 6. Locating Concordance Violations Between Birth Cohorts

Notes: See Figure 5 for data and Appendix J for the test. Each cell plots $\widehat{m}_g/\widehat{s}_g$ on the grid in Table 2, where A is the earlier cohort. Blue (red) cells indicate the later cohort is more (less) mobile at that threshold. Outlined cells are significant at the 5% level for the directional null they violate.

M_λ in Figure 5 is for this comparison and reflects sensitivity to the *average* rank change rather than the tails. Through the lens of Section 5, a crossing of this kind requires non-monotone incidence, e.g., relative skill gains that rose with parental resources across most of the distribution but stalled at the very top.

In contrast, Panel (C) shows that for the 1940s–1950s comparison, although only one order is rejected, it is driven by the rising persistence of high-ranked children. Elsewhere, the 1950s cohort is more mobile in the lower-middle, but not significantly so. Together with Panel (B) this suggests high-rank persistence peaks for the 1950s cohort. Finally, Panel (D) considers a statistically insignificant case where the pattern resembles the crossing copulas, but the violations are small.

Implications for individual measures. Because each cell is a threshold transition-probability comparison, these maps also show which familiar measures rank each pair differently, and why. For the 1950s–1960s, the reversal is confined to the top ventile. Measures that average broadly, or evaluate the copula at quintile thresholds, record falling mobility, while top-ventile persistence records the opposite. Results below show quintile and ventile top persistence rank the cohorts in opposite directions (Corollary E.1). For the 1940s–1950s, measures weighted toward the bottom and middle record rising mobility, measures focused on the top record falling mobility, and M_λ is roughly flat. In the next subsection, I show how this affects measures researchers typically report.

8.4. Do Mobility Measures Move Together?

I now turn to the individual measures researchers typically report. Figure 7 plots the intergenerational elasticity (IGE), the log-income correlation, the rank–rank slope, and directional and trace-based measures from the transition matrix. I normalise all measures so that they increase with mobility (one minus persistence), and defer implementation details to Appendix I.

Panel (A) of Figure 7 replicates the broad patterns in Jácome, Kuziemko, and Naidu (2025):[32] rank persistence and the income correlation fall—thus mobility rises—through the mid-20th century with the exception of a reversal for the 1960s birth cohort, while the IGE rises as income dispersion widens in the latter half of the century. As established in

[32]This replicates results in their Figure 1. However, due to differences in the underlying datasets (see Appendix I) this is best thought of as an independent replication, with modest quantitative differences, rather than a reproduction.

Section 4, this divergence reflects the IGE's sensitivity to changing marginal dispersion.

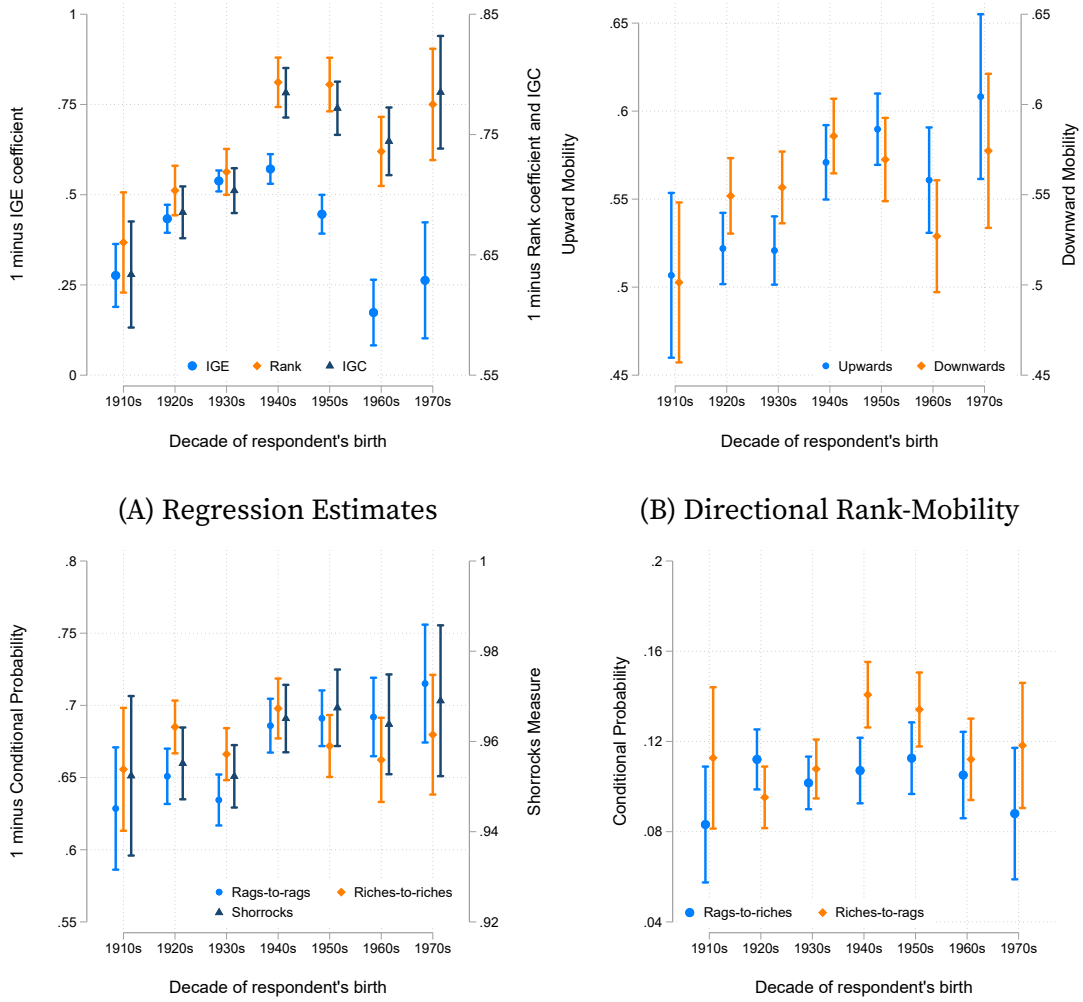
Directional mobility (Panel B), on-diagonal transitions and the Shorrocks trace (Panel C), and off-diagonal transitions (Panel D) largely track the wave–like pattern of the rank–rank slope, even though directional mobility and the Shorrocks trace require stronger assumptions than concordance. Rags-to-rags mobility, however, does not reverse in the 1960s and rags-to-riches mobility falls in the 1970s. These are pointwise movements that are not significant once inference accounts for the full grid of comparisons in Table 2. Where the empirical classification is consistent with concordance—notably the long-run decline from the 1910s to every later cohort—the corner transition probabilities generally move in the same direction, and the exceptions have overlapping confidence intervals. The 1940s–1950s is less clear-cut. In each of Panels (B)–(D), mobility rises at the bottom of the distribution but falls at the top, consistent with Figure 6(C), where violations are significant only among high-ranked children. Conversely, the 1950s–1960s copulas cross. Mobility is not uniformly increasing or decreasing, yet standard approaches to robustness would find nearly all measures fall together.

8.5. Decomposing the Evolution of Mobility

The decomposition in equation (11) allows me to quantify how much of the change in each measure reflects shifts in the dependence structure versus changes in marginal distributions. Figure 8 reports results, highlighting why level–based measures diverge from rank–based measures in practice.

Panel (A) shows the decomposition for rank–rank mobility. For every rank–based measure, all variation comes from the dependence component—marginals are uniform by construction. The concordance tests in Table 2 confirm that this dependence component can itself be partially ordered. Panels (B) and (C) reveal the source of divergence between the IGC and IGE documented in Figure 7. For the IGC, dependence and marginal components move together contributing to rising mobility through mid-century. For the IGE, however, the components move in opposite directions. The dependence component shows rising mobility throughout the century, consistent with rank–based measures, but is increasingly offset by the marginal component.

In this application, the marginal component is driven primarily by the rise in child-to-parent relative income dispersion as the distribution becomes more unequal (see, e.g., Blundell et al. 2018; Piketty, Saez, and Zucman 2018; Guvenen et al. 2022) which stretches children's incomes without reordering them when the copula is unchanged. Rank–based measures track who gains skill, while the IGE also tracks how those gains



(A) Regression Estimates (B) Directional Rank-Mobility

(C) Transition Probabilities (On-Diagonal) (D) Transition Probabilities (Off-Diagonal)

FIGURE 7. Trends in Intergenerational Mobility Measures

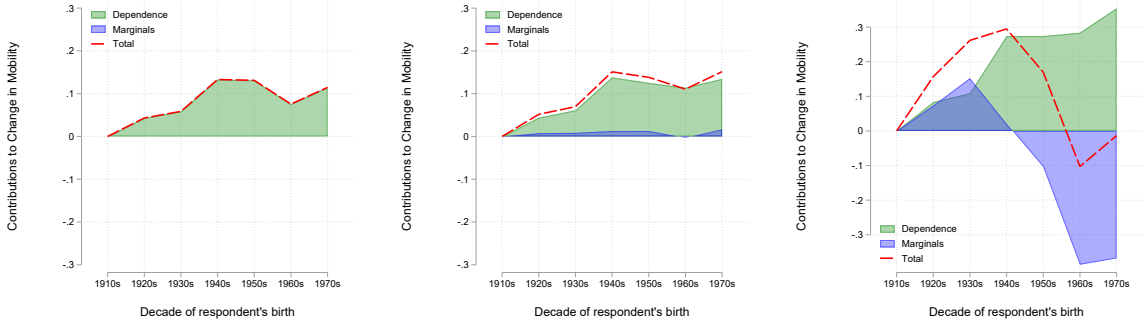
Notes: See Figure 5 for data construction details. Panel (A) reports estimates of IGE, IGC, and rank–rank slope corresponding to Proposition 1. Panel (B) reports estimates of directional rank mobility measures using quintiles ($\Pr(R^K - R^P > 0.2 \mid R^P \leq 0.2)$ and $\Pr(R^K - R^P < -0.2 \mid R^P > 0.8)$) in Proposition 3. Panels (C) and (D) report transition probabilities using quintiles (Proposition 3) and panel (C) additionally reports the Shorrocks Trace Measure (Proposition 4) using deciles. Panels (B)–(D) use 400 bootstrap replications for inference. Panels (A) and (C) normalise measures to increase with mobility (one minus persistence).

are priced in the market.

By the 1970s cohort, the distributional component fully reverses the mobility gains from reduced dependence, producing the decline in IGE-measured mobility. However, between the 1950s and 1960s, the dependence component of the IGE and the rank–rank slope move in opposite directions. With marginals held fixed, this can occur only

because the copulas cross, and it reflects the weight the IGE places on the tails even at fixed marginals, where Figure 6(B) finds a significant reversal.

The decomposition also provides a diagnostic. When the marginal component is large relative to the dependence component, rank–based measures, or an explicit account of distributional change, are needed. Here, the marginal component of the IGE is larger than the dependence component by the 1960s cohort (Panel C), so post-1950 trends in the IGE primarily reflect distributional change.



(A) Rank-Rank Mobility (B) IGC (C) IGE

FIGURE 8. Decomposing Intergenerational Mobility Measures

Notes: See Figure 5 for data construction details. Each panel decomposes the change in mobility (relative to the 1910s cohort) into a dependence (copula) component and a marginal component following equation (11), with Shapley averaging over both orderings to address path dependence. Panel (A) reports the rank–rank slope where the marginal component is zero by construction since rank marginals are uniform under the checkerboard extension. Panel (B) reports the intergenerational correlation (IGC) and Panel (C) reports the intergenerational elasticity (IGE).

Other Measures. Appendix Figure K.4 shows that Fields and Ok (1996) and Cowell and Flachaire (2018) measures display similar overall trends. Moreover, it shows that dependence-driven gains are dominated by changes in the marginal distribution.

9. Discussion and Implications for Applied Researchers

This paper answers two open questions in the literature (e.g., Berman 2022; Deutscher and Mazumder 2023). First, when is one measure sufficient? If distributions are concordance-ordered, any positional mobility measure is an ordinal sufficient statistic for the others. Second, which measure should an applied researcher choose? Concordance eliminates the consequences of measure choice for ordinal comparisons

of positional mobility, without resolving which mobility concept is most appropriate for a particular economic or normative question. When concordance fails, no ranking is independent of the measure: some threshold transition probabilities favour one economy and some the other, so any single summary implicitly decides how to weigh regions where dependence moved in opposite directions. Section 7 provides a structured alternative, reporting local estimates guided by local violations of the concordance order.

Sufficiency also has practical content when data are incomplete. Under concordance, economies can be ranked even when studies do not report the same statistic (one a rank–rank slope, another only a quintile transition matrix) or when only part of the joint distribution is observed and using partial overlap can extend comparisons across many settings. More generally, measure choice becomes a matter of data convenience rather than substance. Transition probabilities require only ordinal ranks and are robust to measurement error common across economies (Corollary 2). The IGE is arguably more exposed to life-cycle bias (Haider and Solon 2006) and top-coding. When only proxies such as education or occupation are available (Ward 2023; Neidhöfer, Serrano, and Gasparini 2018), rank–based measures preserve concordance orderings provided measurement is common across economies (Proposition 5).

Separating dependence from marginal distributions also clarifies a recurring source of empirical disagreement. Rank–based measures isolate positional dependence while the IGE combines it with distributional change. In twentieth-century U.S. data, rank–based measures co–move in a wave–like pattern while the IGE diverges as inequality rises after 1950, reflecting changing marginals that dominate changing positional mobility. Nevertheless, when copulas cross, the IGE and rank–rank correlation diverge even with marginals held fixed.

Finally, the framework applies wherever researchers summarise dependence between relative positions: intragenerational earnings dynamics, wealth mobility, health transmission, or sorting in marriage or labour markets. It extends to trajectories and multiple generations through analogous multivariate orthant orders.

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Online Appendix

A. Measure Agreement for Copula Mobility Measures in Chetty et al. (2014)

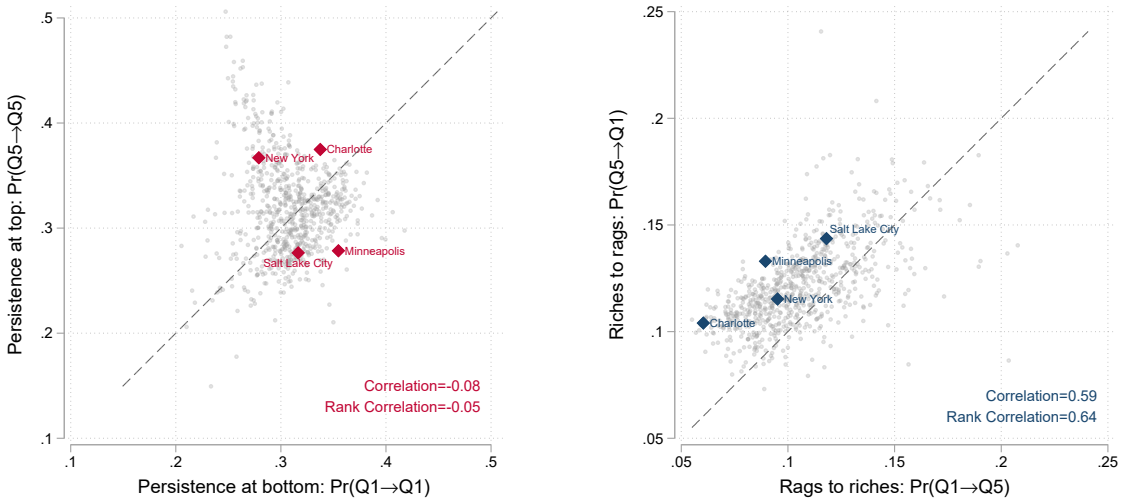
Data and Reconstructing Copulas. I use Commuting Zone Quintile-Quintile Transition Matrices constructed from Chetty et al. (2014)'s published matrices (https://opportunityinsights.org/data/) which provides transition matrices using quintiles of the national income distribution.

To recover the copula on the joint distribution of within CZ ranks, I remap the observed transition matrix (defined over national quintiles, which need not contain equal shares of the subpopulation) onto a uniform copula grid. This isolates mobility's dependence structure from where the subpopulation sits in the national distribution. The remapping assumes that within each national quintile bin, the subpopulation's ranks are uniformly distributed, i.e., the subpopulation's income density within each national quintile mirrors the national density.

This is a natural approximation given data are only observed at the quintile level. However, the population in the top national quintile may be concentrated at the very top of the CZ specific quintile for high income CZs. Given data limitations there are no other alternatives. Importantly, the results in this Appendix are illustrative and do not change either the main empirical application or the theoretical implications arising from the concordance order.

Measure Consensus. Chetty et al. (2014) document substantial heterogeneity in economic mobility across commuting zones (Table III), with the most and least mobile CZ varying depending on the measure used.

Figure A.1 extends the analysis of measure agreement, showing the joint distribution of measures computed from each CZ's copula. This reproduces the results from the introduction for all CZs. Panel A shows how measures of persistence, the probability of remaining in the same quintile as a parent at the bottom and top of the distribution, covary across commuting zones. These measures of intergenerational persistence are negatively correlated, although it is weak, as many CZs are characterised by much more persistence for high-income parents than for low-income parents. Indeed, the



(A) Tail Persistence (B) Tail Mobility

FIGURE A.1. Agreement Among Mobility Measures For Chetty et al. (2014) (All CZs)

Notes: See text for data construction details. Panel (A) reports transition probabilities using quintiles of the within-CZ distribution for the probability of remaining in the bottom and top quintiles across generations. Panel (B) reports probabilities of transitioning from the bottom quintile to the top quintile and vice versa.

implied correlation is negative. Panel B shows the asymmetric transition probabilities for extreme parent–child reversals. These measures of mobility are more positively correlated, but there are many pairs of CZs where the comparison of who is more mobile depends on which of the measures is chosen. While Salt Lake City is more mobile than Charlotte across any of these measures (as well as the rank–rank slope reported in the paper), whether it is more mobile than New York City or if New York City is more mobile than Minneapolis is sensitive to the choice of measure.

Moreover, a broader definition of measure agreement delivers similar results. Among CZs in the top quartile on either persistence measure (panel A), 91.3% are not in the top quartile based on the other measure. Similarly, among CZs in the top quartile on either tail mobility measure (panel B), 61.1% are not in the top quartile based on the other measure.[33]

[33] Mogstad et al. (2024) discuss the inferential problem for rankings based on estimated measures of mobility. However, the pairwise comparisons above do not have the same concern.

B. Discussion for Section 2

B.1. Mass-Points and Tie-Breaking Ranks

In the main body, I present the simple definition of a rank, $R^K = F_K(Y^K)$ and $R^P = F_P(Y^P)$, which are uniformly distributed on $[0, 1]$ and map from incomes to the unit interval when the marginal distribution is continuous. I now turn to accounting for discrete values and mass points.

Let

$$F_s(y-) \equiv \Pr(Y^s < y) \equiv \lim_{x \uparrow y} F_s(x),$$

define the left limit of the marginal CDF at y for $s \in \{K, P\}$. Then ranks are given by their percentile within the marginal distribution, with random tie-breaking within mass points. Formally,

$$R^s = F_s(Y^s-) + V^s[F_s(Y^s) - F_s(Y^s-)] \; s \in \{K, P\},$$

which are uniformly distributed on $[0, 1]$, map from incomes to the unit interval, and collapse to the familiar rank, $R^K = F_K(Y^K)$ and $R^P = F_P(Y^P)$, when the marginal distribution is continuous.

B.2. Copula Uniqueness

Sklar's Theorem (1959) states that the copula associated with a joint distribution is unique if and only if marginal distributions are continuous. When the random variables take on discrete values or are drawn from distributions with mass points, the copula is generically not uniquely determined. Instead, it is only unique on the set of ranks the marginals actually take (Thm 2.3.3, Nelsen 2006). There are infinitely many valid extensions for values outside the observed support of ranks.

Thus, even if mass points are only at the boundary of support (as in the case of censored or truncated income observations), the copula is not unique. The results in this paper do not, unless stated otherwise, rely on continuity or differentiability of the copula. Uniqueness is restored by adopting the checkerboard convention.

Addressing this indeterminacy is referred to as an *extension* of the copula. Unfortunately, extensions are not necessarily guaranteed to preserve the concordance order. Thus, the choice of extension is not without loss. There are three solutions to this problem for a given empirical copula:

(i) Use a checkerboard extension (Genest and Nešlehová 2007) which is piecewise bilinear

(ii) Consider Carley (2002) bounds on copulas

(iii) Approximate the discrete outcomes with a specific parametric copula family.

Solution (iii) is the easiest solution, but also requires the strongest assumptions making it typically unappealing. Solution (ii) would replace inequalities on the copula required for the concordance order with worst-case inequalities comparing upper and lower bounds. This is extremely flexible, but can be limiting when the bounds are not sharp. Finally, solution (i) restricts the set of admissible extensions and selects a bilinear interpolation on the density. This guarantees uniqueness and, as shown by Propositions 11 and 13 in Genest and Nešlehová (2007), the checkerboard extension preserves the concordance order of the original joint-distributions. Consequently, I assume that discreteness is addressed through checkerboard extension of the copula. While this is a mild assumption, identical results can be optained for parametric copulas and bounds are available in the other case.

C. Proofs for Section 3

Proof of Proposition 1. The population value of β is given by $cov(\ln y_i^k, \ln y_i^p)/var(\ln y_i^p)$. The denominator is constant and the numerator is concordance–increasing as a direct consequence of Lemma 3 in Lehmann (1966) (see also Lemma 2, Hoeffding 1940). Similarly, the population value of ρ is concordance–increasing from Corollary 3.2 in Tchen (1980). □

Proof of Proposition 2. Let $U = R^K$ and $V = R^P$. Since ranks are uniformly distributed, $\Pr(V \leq r) = r$. Moreover,

$$\begin{aligned} E\left[U \mathbf{1}_{V \leq r\}}\right] &= \int_0^1 \Pr(U > u, V \leq r)\, du \\ &= r - \int_0^1 C(u, r)\, du. \end{aligned} \tag{C.1}$$

It follows that

$$E\left[U \mid V \leq r\right] = 1 - \frac{1}{r} \int_0^1 C(u, r)\, du. \tag{C.2}$$

Therefore, $C^A \succeq C^B$ implies

$$E^A\left[U \mid V \leq r\right] \leq E^B\left[U \mid V \leq r\right] \qquad \forall\, r \in (0,1). \tag{C.3}$$

Similarly, using $E[U] = 1/2$,

$$E\left[U \mid V > r\right] = \frac{\frac{1}{2} - r + \int_0^1 C(u,r)\,du}{1-r}. \tag{C.4}$$

Hence $C^A \succeq C^B$ implies

$$E^A\left[U \mid V > r\right] \geq E^B\left[U \mid V > r\right] \qquad \forall\, r \in (0,1), \tag{C.5}$$

which proves the result. □

Proof of Corollary 1. The fitted conditional expected rank is

$$E^i\left[R^K \mid R^P = r\right] = \frac{1}{2} + \rho_i\left(r - \frac{1}{2}\right), \qquad i \in \{A, B\}, \tag{C.6}$$

where the intercept follows from the uniformity of ranks. Proposition 1 implies $\rho_A \geq \rho_B$ when $C^A \succeq C^B$. The conditional expected rank is therefore decreasing relative to the concordance ordering below the median and increasing relative to the concordance ordering above the median. □

Proof of Corollary C.1. Where the relevant derivative exists, let

$$m_i(r) = E^i\left[R^K \mid R^P = r\right]. \tag{C.7}$$

The conditional expected rank satisfies

$$m_i(r) = 1 - \int_0^1 \partial_r C^i(u,r)\,du. \tag{C.8}$$

It follows from equation (C.38) that

$$m_A(r) - m_B(r) = -G'(r). \tag{C.9}$$

Since G is single-peaked at $r^\star$, $G'(r) \geq 0$ below $r^\star$ and $G'(r) \leq 0$ above $r^\star$. Hence

$$m_A(r) \leq m_B(r) \quad \text{for } r \leq r^\star, \qquad m_A(r) \geq m_B(r) \quad \text{for } r \geq r^\star,$$

which establishes the result. □

Proof of Proposition 3. The transition probabilities can be rewritten using the conditional CDF. For the positive diagonal case we have

$$TP\left[R^K \le \tau^K \mid R^P \le \tau^P\right] = \frac{\Pr(R^K \le \tau^K \cap R^P \le \tau^P)}{\Pr(R^P \le \tau^P)} = \frac{C(\tau^K, \tau^P)}{\tau^P}, \qquad \tau^P > 0. \tag{C.10}$$

and

$$TP\left[R^K > \tau^K \mid R^P > \tau^P\right] = \frac{\Pr(R^K > \tau^K \cap R^P > \tau^P)}{\Pr(R^P > \tau^P)} = \frac{1 - \tau^P - \tau^K + C(\tau^K, \tau^P)}{1 - \tau^P}, \qquad \tau^P < 1. \tag{C.11}$$

For the lower-left transition probability, the inequality in the proposition is satisfied for all thresholds iff

$$C^A(\tau^K, \tau^P) \ge C^B(\tau^K, \tau^P) \quad \forall\, \tau^K \in [0,1],\ \tau^P \in (0,1], \tag{C.12}$$

which, together with the copula boundary conditions, is equivalent to the definition of increased concordance. This also establishes the converse in part (iii). For the off-diagonal case we have

$$TP\left[R^K > \tau^K \mid R^P \le \tau^P\right] = 1 - TP\left[R^K \le \tau^K \mid R^P \le \tau^P\right], \tag{C.13}$$

and

$$TP\left[R^K \le \tau^K \mid R^P > \tau^P\right] = 1 - TP\left[R^K > \tau^K \mid R^P > \tau^P\right], \tag{C.14}$$

with the stated inequality following directly from the results established in the positive diagonal case. Turning to directional rank mobility, fix $\tau \in (0, r^\star]$ and $s \in [0,1]$. By the definition of conditional probability and the uniformity of parental ranks,

$$\begin{aligned} URM(s,\tau) &= \Pr\left(R^K - R^P > s \mid R^P \le \tau\right) \\ &= \frac{\Pr\left(R^K > R^P + s,\ R^P \le \tau\right)}{\Pr(R^P \le \tau)} \\ &= \frac{1}{\tau}\int_0^\tau \Pr\left(R^K > r + s \mid R^P = r\right)\, dr \\ &= \frac{1}{\tau}\int_0^\tau \left[1 - C_{K|P}(r + s \mid r)\right]\, dr, \end{aligned} \tag{C.15}$$

where $C_{K|P}(\cdot \mid r)$ denotes the regular conditional distribution of child rank given parental

rank r, and is understood to equal one whenever $r + s \geq 1$.

For $\tau \leq r^\star$, Assumption 1i implies, for almost every $r \in [0, \tau]$,

$$C^A_{K|P}(r + s \mid r) \geq C^B_{K|P}(r + s \mid r). \tag{C.16}$$

Equivalently,

$$Pr_A\left(R^K > r + s \mid R^P = r\right) \leq Pr_B\left(R^K > r + s \mid R^P = r\right) \tag{C.17}$$

almost everywhere on $[0, \tau]$. Integrating over parental ranks therefore gives

$$URM^A(s, \tau) \leq URM^B(s, \tau) \qquad \forall\, s \in [0, 1], \quad \tau \in (0, r^\star], \tag{C.18}$$

which establishes the upward rank mobility result.

The proof for downward rank mobility is analogous. Fix $\tau \in [r^\star, 1)$ and $s \in [0, 1]$. Again using the uniformity of parental ranks,

$$\begin{aligned} DRM(s, \tau) &= \Pr\left(R^P - R^K > s \mid R^P \geq \tau\right) \\ &= \frac{\Pr\left(R^K < R^P - s,\ R^P \geq \tau\right)}{\Pr(R^P \geq \tau)} \\ &= \frac{1}{1-\tau}\int_\tau^1 \Pr\left(R^K < r - s \mid R^P = r\right)\, dr \\ &= \frac{1}{1-\tau}\int_\tau^1 C_{K|P}(r - s \mid r)\, dr, \end{aligned} \tag{C.19}$$

where the conditional CDF is understood to equal zero whenever $r - s \leq 0$. For $\tau \geq r^\star$, Assumption 1i implies, for almost every $r \in [\tau, 1]$,

$$C^A_{K|P}(r - s \mid r) \leq C^B_{K|P}(r - s \mid r). \tag{C.20}$$

Hence,

$$DRM^A(s, \tau) \leq DRM^B(s, \tau) \qquad \forall\, s \in [0, 1], \quad \tau \in [r^\star, 1), \tag{C.21}$$

which establishes the downward rank mobility result.

□

Proof of Proposition 4. Fields and Ok Measure. The aggregate income movement per

capita is

$$M_{FO} = \mathbb{E}\left[\left|Y^K - Y^P\right|\right]. \tag{C.22}$$

The function $\psi(x, y) = |x - y|$ is submodular (equivalently, it is a convex function of the difference $x - y$). Hence, since concordance is equivalent to the supermodular order under common marginals,

$$C^A \succeq C^B \quad \Longrightarrow \quad \mathbb{E}_A\left[\left|Y^K - Y^P\right|\right] \leq \mathbb{E}_B\left[\left|Y^K - Y^P\right|\right]. \tag{C.23}$$

This also follows directly from Theorem 9.A.18 in Shaked and Shanthikumar (2007) or Corollary 2.3 (and Example 2) in Tchen (1980). The proof for the log measure (Fields and Ok 1999) is identical after replacing Y^s by $\log Y^s$.

The Fields and Ok (1996, 1999) measure can be decomposed into structural mobility,

$$\left|\mathbb{E}[Y^K] - \mathbb{E}[Y^P]\right|, \tag{C.24}$$

and an exchange component,

$$2\,\mathbb{E}\left[|Y^K - Y^P|\mathbf{1}\{(Y^K, Y^P) \in S\}\right], \tag{C.25}$$

where

$$S = \begin{cases} \{(Y^K, Y^P) : Y^K < Y^P\}, & \text{if } \mathbb{E}[Y^K] \geq \mathbb{E}[Y^P], \\ \{(Y^K, Y^P) : Y^K > Y^P\}, & \text{if } \mathbb{E}[Y^K] < \mathbb{E}[Y^P]. \end{cases} \tag{C.26}$$

The structural component depends only on the marginals and is therefore identical under C^A and C^B. Consequently, the exchange component inherits the same ordering as total mobility and is decreasing in the concordance order.

Cowell and Flachaire Measure. The aggregate mobility measure is

$$M_{CF,\alpha} = \mathbb{E}\left[\psi_\alpha(Y^K, Y^P)\right], \tag{C.27}$$

where

$$\psi_\alpha(x, y) = \frac{1}{\alpha(\alpha - 1)}\left(\frac{x}{\bar{x}}\right)^\alpha \left(\frac{y}{\bar{y}}\right)^{1-\alpha}, \tag{C.28}$$

and $\bar{x} = \mathbb{E}[Y^K]$ and $\bar{y} = \mathbb{E}[Y^P]$. On its domain,

$$\frac{\partial^2 \psi_\alpha(x, y)}{\partial x\, \partial y} = -\frac{1}{\bar{x}^\alpha \bar{y}^{1-\alpha}} x^{\alpha-1} y^{-\alpha} \leq 0. \tag{C.29}$$

Thus ψ_α is submodular. Since the marginals, and therefore $\overline{x}$ and $\overline{y}$, are common across economies, concordance (equivalently, the supermodular order under common marginals) implies

$$C^A \succeq C^B \quad \Longrightarrow \quad M^A_{CF,\alpha} \leq M^B_{CF,\alpha}. \tag{C.30}$$

Shorrocks Measure. Let

$$I_i = \left(\frac{i-1}{q}, \frac{i}{q}\right], \qquad i = 1, \ldots, q.$$

The elements of the transition matrix $\mathcal{Q}$ are

$$\mathcal{Q}_{i,j} = \Pr\left(R^K \in I_j \mid R^P \in I_i\right) = q\left(C\left(\frac{i}{q}, \frac{j}{q}\right) - C\left(\frac{i-1}{q}, \frac{j}{q}\right) - C\left(\frac{i}{q}, \frac{j-1}{q}\right) + C\left(\frac{i-1}{q}, \frac{j-1}{q}\right)\right). \tag{C.31}$$

Since parental ranks are uniformly distributed, $\Pr(R^P \in I_i) = 1/q$, and hence

$$\mathcal{Q}_{i,j} = q \Pr\left(R^K \in I_j,\ R^P \in I_i\right). \tag{C.32}$$

In particular, the diagonal elements satisfy

$$\mathcal{Q}_{i,i} = q \Pr\left(R^K \in I_i,\ R^P \in I_i\right). \tag{C.33}$$

Assumption 1ii therefore implies

$$\mathcal{Q}^A_{i,i} \geq \mathcal{Q}^B_{i,i} \qquad \forall\, i, \tag{C.34}$$

and consequently

$$\operatorname{tr}\left(\mathcal{Q}^A\right) \geq \operatorname{tr}\left(\mathcal{Q}^B\right). \tag{C.35}$$

Since the Shorrocks mobility measure,

$$M = \frac{q - \operatorname{tr}(\mathcal{Q})}{q-1}, \tag{C.36}$$

is decreasing in the trace,

$$M^A \leq M^B. \tag{C.37}$$

The proof extends immediately to transition matrices defined using arbitrary common rank cut-points. □

Proof of Proposition 5. This is immediate as an application of the closure properties of the concordance order. See Theorem 9.A.1 Shaked and Shanthikumar (2007).

$$(E_A^K, E_A^P) \succeq_c (E_B^K, E_B^P).$$

Moreover, since ranks have common uniform marginals across economies, and the functions H^s and distributions of ε^s are common across economies, the proxy marginals are also common. Hence the corresponding checkerboard copulas inherit the same concordance ordering, $\widetilde{C}_E^A \succeq \widetilde{C}_E^B$. □

Proof of Corollary 2. The result follows immediately from the closure properties of the concordance order (see Theorem 9.A.1 Shaked and Shanthikumar 2007). □

C.1. Additional Results

More generally, the pointwise conditional expectation in corollary 1 need not be linear. A similar rotation result obtains under a weaker shape restriction on the difference in the integrated copulas. Define

$$G(r) = \int_0^1 \left[C^A(u, r) - C^B(u, r) \right] du. \tag{C.38}$$

COROLLARY C.1 (Pointwise Conditional Expected Rank Measures). *Suppose the copulas are continuously differentiable in parental rank and $G(r)$ is single-peaked at $r^\star$, so that $G'(r) \geq 0$ for $r < r^\star$ and $G'(r) \leq 0$ for $r > r^\star$. If $C^A \succeq C^B$, the conditional expected rank curve rotates around $r^\star$:*

i. it is concordance decreasing below $r^\star$, $\mathbb{E}^A \left[R^K \mid R^P = r \right] \leq \mathbb{E}^B \left[R^K \mid R^P = r \right] \quad \forall\, r \leq r^\star$, and

ii. it is concordance–increasing above $r^\star$, $\mathbb{E}^A \left[R^K \mid R^P = r \right] \geq \mathbb{E}^B \left[R^K \mid R^P = r \right] \quad \forall\, r \geq r^\star$.

D. Additional Discussion Section 4: Illustration of the Decomposition

The log-normal case makes the decomposition in equation (10) explicit. Suppose parent and child incomes are jointly log-normal:

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N \left[\begin{pmatrix} \mu_K \\ \mu_P \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho\sigma_K\sigma_P \\ \rho\sigma_K\sigma_P & \sigma_P^2 \end{pmatrix} \right].$$

The copula is Gaussian with parameter ρ with marginals determined by (μ_K, σ_K) and (μ_P, σ_P). For the rank–rank correlation, $\mathrm{Corr}(R^K, R^P) = \frac{6}{\pi}\arcsin(\rho/2) \approx \rho$ (Kruskal 1958), so the marginal component in (10) is exactly zero—changes in σ_K or σ_P have no effect. For the IGE, $\beta = \rho \cdot \sigma_K/\sigma_P$, and the decomposition yields:

$$\Delta\beta = \underbrace{(\rho_B - \rho_A) \cdot \frac{\sigma_K^B}{\sigma_P^B}}_{\text{Copula component}} + \underbrace{\rho_A \cdot \left(\frac{\sigma_K^B}{\sigma_P^B} - \frac{\sigma_K^A}{\sigma_P^A}\right)}_{\text{Marginal component}}.$$

The copula component is ordered by concordance: higher ρ implies higher β, holding marginals fixed. But the marginal component can offset or even reverse this ordering. An economy with lower concordance ($\rho_B < \rho_A$) can exhibit a *higher* IGE, thus *lower mobility*, if the ratio of child-to-parent dispersion rises sufficiently ($\sigma_K^B/\sigma_P^B \gg \sigma_K^A/\sigma_P^A$).

E. Proofs for Section 6

Proof of Proposition 7. Since $C^A \succeq C^\dagger$ and M is assumed monotone in the concordance order we have $M(C^A) \leq M(C^\dagger)$. By uniform continuity of M we have

$$|M(C^\dagger) - M(C^B)| \leq \omega_M(\|C^\dagger - C^B\|_\infty) \leq \omega_M(\epsilon). \tag{E.1}$$

Therefore $M(C^A) \leq M(C^\dagger) \leq M(C^B) + \omega_M(\epsilon)$ which proves the first result. If M is Lipschitz continuous with constant L_M, then $\omega_M(\epsilon) \leq L_M\epsilon$. □

E.1. Additional Results

Concordance violations can manifest in ranking reversals even for pre-specified discretisations including standard empirical objects (e.g., quintile, decile, or ventile transition matrices). The following corollary gives sufficient conditions for such reversals.

COROLLARY E.1 (Ranking Reversals at Standard Discretisations). *Let C^A and C^B be copulas with densities c^A and c^B such that $\Delta C \equiv C^A - C^B$ changes sign. The conditions below can equivalently be stated in terms of the copula measure of ΔC , but the density formulation is more transparent. Write $\Delta c \equiv c^A - c^B$. Fix two reporting grids*

$$\Pi_1 = \{0 = t_0^1 < t_1^1 < \cdots < t_K^1 = 1\}, \quad \Pi_2 = \{0 = t_0^2 < t_1^2 < \cdots < t_J^2 = 1\}$$

e.g., quintiles and deciles. Define cells $R^1_{ij} = [t^1_{i-1}, t^1_i] \times [t^1_{j-1}, t^1_j]$ *and* $R^2_{kl} = [t^2_{k-1}, t^2_k] \times [t^2_{l-1}, t^2_l]$.

If there exist cells such that:

i. Δc *maintains constant sign on each cell, and*

ii. $\int\int_{R^1_{i,j}} \Delta c(u,v)\, du\, dv > 0$, *but* $\int\int_{R^2_{k,l}} \Delta c(u,v)\, du\, dv < 0$,

then the corresponding transition probabilities reverse rankings across the two grids whenever the detecting cells have the same mobility orientation.

When concordance holds, all standard measures considered here agree. When the copulas are not concordance ordered, uniform agreement is impossible: Proposition 3 implies that threshold transition probabilities must disagree somewhere. Researchers cannot avoid these disagreements by restricting attention to "simpler" or more "transparent" measures as under the conditions of Corollary E.1, these reversals appear even among transition probabilities reported on the standard grids routinely used in empirical work. A researcher using quintile transition matrices may conclude economy A has lower mobility, while another using decile matrices concludes A has higher mobility. This is not measurement error, but reflects genuine ambiguity in how to weight regions where parent–child dependence differs in opposite directions.

F. Additional Results for Section 7: Robust Measure Choice

Proposition 7 shows concordance results are not a knife-edge property: if two copulas are within ϵ of satisfying concordance (in sup-norm), measures are approximately ordered, with error bounded by ϵ times a measure-specific constant. This matters empirically because real-world copulas may nearly—but not exactly—satisfy concordance due to sampling variation or genuine but minor departures.

The bound L_M varies across measures and determines their sensitivity to concordance violations. Measures that average dependence broadly—such as the rank–rank slope, with $L_\rho = 12$—have smaller Lipschitz constants and degrade smoothly. Measures that concentrate on narrow regions, such as transition probabilities conditioning on rare events, can have larger constants and are more sensitive. This explains why rank–based measures exhibit tighter co-movement in empirical work (Deutscher and Mazumder 2023): their smaller Lipschitz constants make them less sensitive to the minor concordance violations inevitably present in real data. Approximate concordance expands empirical relevance by allowing researchers to test whether copulas satisfy concordance within estimation error. This accommodates realistic settings where economies exhibit small departures due to institutional details

or demographics while mobility measures still move together. I now turn to discussing measure selection motivated by robustness to approximate concordance.

A similar approach characterizes the sensitivity of measures to these deviations. Recall from equation (16) that any twice-differentiable measure M satisfies[34]

$$M_M^A - M_M^B = \int_0^1 \int_0^1 W_M(R^K, R^P) \left[C^A(R^K, R^P) - C^B(R^K, R^P) \right] dR^K \, dR^P, \tag{F.1}$$

where W_M is the measure specific weighting-kernel with Lipschitz constant:

$$L_M = |W_M|_1 \equiv \int_0^1 \int_0^1 |W_M(u, v)| du \, dv \, . \tag{F.2}$$

COROLLARY F.1 (Optimal Robust Measure). *If the true copula C satisfies $||C - C^\dagger||_\infty \leq \epsilon$, then $M(C) \in [M(C^\dagger) - L_M \epsilon, M(C^\dagger) + L_M \epsilon]$. Among concordance-monotone measures, an ϵ-optimal choice of measure minimizes L_M.*

Interpretation. Measures concentrating weight in specific regions (e.g., the intergenerational elasticity or local conditional mobility) have larger L_M and greater sensitivity to concordance violations as do those measures expressed in larger units. Measures spreading weight uniformly (e.g., coarse transitions or rank–rank slopes)—have smaller L_M and are robust to approximate concordance.

F.1. Lipschitz Constants for Robust Measure Choice

Throughout, let C^A, C^B be copulas on $[0, 1]^2$ with the metric $\|C^A - C^B\|_\infty \equiv \sup_{(u,v)\in[0,1]^2} |C^A(u, v) - C^B(u, v)|$. For any functional M that admits the representation in (F.3),

$$M(C^A) - M(C^B) = \int_0^1 \int_0^1 W_M(u, v) \left(C^A(u, v) - C^B(u, v) \right) du \, dv, \qquad L_M = |W_M|_1, \tag{F.3}$$

the Lipschitz bound follows immediately:

$$|M(C^A) - M(C^B)| \leq |W_M|_1 \, \|C^A - C^B\|_\infty. \tag{F.4}$$

To maintain simple exposition I focus on cases where the measure is twice-differentiable or can be bounded directly.

[34] Analogous results apply to discrete cases, but the differentiable case simplifies exposition.

Measure (Table 1)	L_M **for** $\|C^A - C^B\|_\infty$	**Conditions / notes**
IGE (β)	$L_\beta = \dfrac{(\overline{x}^K - \underline{x}^K)(\overline{x}^P - \underline{x}^P)}{\mathrm{Var}(X^P)}$	Fixed marginals; $X^s = \ln Y^s$; bounded log-income support $X^s \in [\underline{x}^s, \overline{x}^s]$. Unbounded support requires additional tail restrictions.
Rank–rank correlation (ρ)	$L_\rho = 12$	Uniform rank marginals (Spearman/Hoeffding representation).
Projected CER	$L_{\mathrm{CER}(r)} = 12\left\|r - \frac{1}{2}\right\|$	Parametric-only: $\mathrm{CER}_r = \frac{1}{2} + \rho(r - \frac{1}{2})$.
TP	$L = \dfrac{1}{\tau_p}$ for $R^P \le \tau_p$; $L = \dfrac{1}{1-\tau_p}$ for $R^P > \tau_p$	Requires $\tau_p \in (0,1)$. Direct evaluation bound from copula at (τ_k, τ_p).
Shorrocks	$L_{\mathrm{Sh}} \le \dfrac{4q^2}{q-1}$	Discrete binning (q groups).
Cowell & Flachaire (α)	Levels: $\dfrac{(b_K^\alpha - a_K^\alpha)(b_P^{1-\alpha} - a_P^{1-\alpha})}{\alpha(1-\alpha)(\overline{Y}^K)^\alpha(\overline{Y}^P)^{1-\alpha}}$ Ranks: $\dfrac{2}{\alpha(1-\alpha)}$, $\alpha \in (0,1)$	Levels: bounded positive supports $[a_K, b_K]$ and $[a_P, b_P]$. Ranks: uniform marginals.

TABLE F.1. Simplified summary of Lipschitz constants used (or not available) for the robust selection rule under the copula sup-norm.

F.1.1. IGE (β)

Let $X^K = \ln Y^K$ and $X^P = \ln Y^P$, with fixed marginal distributions F_{X^K} and F_{X^P} and corresponding densities f_{X^K} and f_{X^P}. The intergenerational elasticity is

$$\beta(C) = \frac{Cov(X^K, X^P)}{Var(X^P)},$$

where $Var(X^P)$ is constant under fixed marginals. By Hoeffding's identity and Sklar's theorem,

$$\begin{aligned} Cov(X^K, X^P) &= \int\int \left(F_{X^K, X^P}(x^K, x^P) - F_{X^K}(x^K) F_{X^P}(x^P) \right) dx^K \, dx^P \\ &= \int_0^1 \int_0^1 \frac{C(u,v) - uv}{f_{X^K}\left(F_{X^K}^{-1}(u)\right) \, f_{X^P}\left(F_{X^P}^{-1}(v)\right)} \, du \, dv. \end{aligned} \tag{F.5}$$

Hence, in the notation of equation (16), the weighting kernel for the IGE is

$$W_\beta(u,v) = \frac{1}{Var(X^P)} \frac{1}{f_{X^K}\left(F^{-1}_{X^K}(u)\right)\, f_{X^P}\left(F^{-1}_{X^P}(v)\right)}. \tag{F.6}$$

It follows that, on bounded support for log-incomes,

$$X^K \in [\underline{x}^K, \overline{x}^K], \qquad X^P \in [\underline{x}^P, \overline{x}^P],$$

the IGE is Lipschitz in the sup-norm with constant

$$\begin{aligned} L_\beta &= |W_\beta|_1 \\ &= \frac{1}{Var(X^P)} \int_0^1 \int_0^1 \frac{1}{f_{X^K}\left(F^{-1}_{X^K}(u)\right)\, f_{X^P}\left(F^{-1}_{X^P}(v)\right)}\, du\, dv \\ &= \frac{1}{Var(X^P)} \left(\int_{\underline{x}^K}^{\overline{x}^K} dx^K \right) \left(\int_{\underline{x}^P}^{\overline{x}^P} dx^P \right) \\ &= \frac{(\overline{x}^K - \underline{x}^K)(\overline{x}^P - \underline{x}^P)}{Var(X^P)}. \end{aligned} \tag{F.7}$$

Therefore,

$$|\beta(C^A) - \beta(C^B)| \le L_\beta \, \|C^A - C^B\|_\infty.$$

The same argument applies on any trimmed support of log-incomes. When log-incomes are unbounded, a finite global Lipschitz constant need not exist without additional tail restrictions.

F.1.2. Rank–rank slope (ρ)

With uniform rank marginals, the rank–rank slope is $\rho = \mathrm{Cov}(U,V)/\mathrm{Var}(V)$, and $\mathrm{Var}(V) = 1/12$. Hoeffding's identity gives $\mathrm{Cov}(U,V) = \int_0^1\int_0^1 (C(u,v) - uv)\, du\, dv$, so

$$\rho(C) = 12 \int_0^1 \int_0^1 \big(C(u,v) - uv\big)\, du\, dv.$$

Thus

$$\rho(C^A) - \rho(C^B) = 12 \int_0^1 \int_0^1 \big(C^A(u,v) - C^B(u,v)\big)\, du\, dv,$$

and

$$|\rho(C^A) - \rho(C^B)| \le 12 \iint |C^A - C^B| \le 12 \, \|C^A - C^B\|_\infty.$$

Hence

$$L_\rho = 12.$$

F.1.3. CER

In the projection case the conditional expected rank at parent rank r is

$$\mathrm{CER}_r(C) = \tfrac{1}{2} + \rho(C)\,(r - \tfrac{1}{2}),$$

so Lipschitz continuity follows by composition with ρ:

$$|\mathrm{CER}_r(C^A) - \mathrm{CER}_r(C^B)| = |r - \tfrac{1}{2}|\,|\rho(C^A) - \rho(C^B)| \le 12|r - \tfrac{1}{2}|\,\|C^A - C^B\|_\infty.$$

Therefore

$$L_{\mathrm{CER}(r)} = 12|r - \tfrac{1}{2}|.$$

F.1.4. TP (transition probabilities)

Appendix C rewrites diagonal transition probabilities as evaluations of C:

$$\mathrm{TP}\big[R^K \le \tau_k \,\big|\, R^P \le \tau_p\big] = \frac{C(\tau_k, \tau_p)}{\tau_p}, \qquad \tau_p \in (0, 1).$$

Hence

$$\Big|\mathrm{TP}_A - \mathrm{TP}_B\Big| = \frac{|C^A(\tau_k, \tau_p) - C^B(\tau_k, \tau_p)|}{\tau_p} \le \frac{1}{\tau_p}\,\|C^A - C^B\|_\infty,$$

so

$$L_{\mathrm{TP}\le} = 1/\tau_p.$$

Similarly, for upper-tail conditioning

$$\mathrm{TP}\big[R^K > \tau_k \,\big|\, R^P > \tau_p\big] = \frac{1 - \tau_p - \tau_k + C(\tau_k, \tau_p)}{1 - \tau_p},$$

and the same argument yields

$$L_{\mathrm{TP}>} = 1/(1 - \tau_p).$$

F.1.5. Shorrocks (trace measure)

For q bins the discretized transition matrix is

$$Q_{ij}(C) = q\Big(C(\tfrac{i}{q}, \tfrac{j}{q}) - C(\tfrac{i-1}{q}, \tfrac{j}{q}) - C(\tfrac{i}{q}, \tfrac{j-1}{q}) + C(\tfrac{i-1}{q}, \tfrac{j-1}{q})\Big),$$

and $\mathrm{trace}(Q) = \sum_{i=1}^{q} Q_{ii}$. Let $\Delta C = C^A - C^B$. Then each diagonal increment is a signed sum of four evaluations of ΔC, so

$$|Q_{ii}(C^A) - Q_{ii}(C^B)| \le 4q\,\|\Delta C\|_\infty.$$

Summing over i yields

$$|\mathrm{trace}(Q(C^A)) - \mathrm{trace}(Q(C^B))| \le 4q^2\,\|C^A - C^B\|_\infty.$$

For Shorrocks' trace index $\mathrm{Sh}(Q) = (q - \mathrm{trace}(Q))/(q-1)$, it follows that

$$|\mathrm{Sh}(C^A) - \mathrm{Sh}(C^B)| \le \frac{4q^2}{q-1}\,\|C^A - C^B\|_\infty, \qquad L_{\mathrm{Sh}} \le \frac{4q^2}{q-1}.$$

F.1.6. Cowell & Flachaire (parameter α)

From equation (C.29), the mobility intensity satisfies

$$\psi_{\alpha,12}(x, y) = -\frac{x^{\alpha-1}\, y^{-\alpha}}{(\bar{Y}^K)^{\alpha}(\bar{Y}^P)^{1-\alpha}}.$$

For continuous marginals with strictly positive densities, equation (15) gives

$$W_{\mathrm{CF}}(u, v) = \frac{\psi_{\alpha,12}(F_K^{-1}(u), F_P^{-1}(v))}{f_K(F_K^{-1}(u))\, f_P(F_P^{-1}(v))}.$$

Thus, by a change of variables,

$$\begin{aligned} L_{\mathrm{CF}} &= \|W_{\mathrm{CF}}\|_1 \\ &= \frac{1}{(\bar{Y}^K)^{\alpha}(\bar{Y}^P)^{1-\alpha}} \left(\int_{a_K}^{b_K} x^{\alpha-1}\,dx\right)\left(\int_{a_P}^{b_P} y^{-\alpha}\,dy\right) \\ &= \frac{(b_K^{\alpha} - a_K^{\alpha})(b_P^{1-\alpha} - a_P^{1-\alpha})}{\alpha(1-\alpha)(\bar{Y}^K)^{\alpha}(\bar{Y}^P)^{1-\alpha}}, \end{aligned}$$

for $\alpha \in (0,1)$ and bounded positive supports $[a_K, b_K]$ and $[a_P, b_P]$.

The same bound extends to distributions with mass points: integration by parts in income levels yields

$$M_{\mathrm{CF}}(C^A) - M_{\mathrm{CF}}(C^B) = \iint \psi_{\alpha,12}(x,\, y) \left[C^A(F_K(x), F_P(y)) - C^B(F_K(x), F_P(y)) \right] dx\, d\, y,$$

which is bounded by $L_{\mathrm{CF}} \| C^A - C^B \|_\infty$.

When evaluated on ranks, both marginals are uniform on $[0,1]$ with means $1/2$, giving

$$L_{\mathrm{CF}} = \frac{2}{\alpha(1-\alpha)}.$$

G. Additional Results for Section 7: Models of Scientific Communication

The reporting protocol in Section 7.2 can be understood through the lens of Andrews and Shapiro (2021), who study when a statistical report is adequate for a heterogeneous audience.

In their framework, an analyst makes a report to an audience after observing data. Audience members differ in their objectives and may therefore act differently following a given report. A report is *communication-admissible* if no alternative report would make all audience members weakly better off and at least one strictly better off.

The mobility setting maps directly into this framework. The analyst is a researcher comparing two economies. The audience consists of readers who care about different mobility measures—one cares about the IGE, another about bottom-quintile transitions, a third about the Shorrocks trace. Formally, each audience member m has a weighting kernel W_m and evaluates the comparison through the integral $\int \int W_m \cdot \Delta C$. The researcher's report is whatever summary statistics they publish.

When is single–measure reporting admissible?. When concordance holds, any single concordance-monotone measure determines the sign of all others. A report consisting of the global rank–rank slope alone is communication-admissible: every audience member can determine whether their preferred measure ranks A above or below B.

When concordance fails, the global rank–rank slope is communication-inadmissible. The argument follows directly from Corollary E.1. There exist audience members—specifically, readers whose preferred measures are transition probabilities at different

thresholds—who reach opposite conclusions. The global slope integrates ΔC against a uniform kernel, reflecting one net direction. A reader whose preferred measure loads on a region where ΔC takes the opposite sign is strictly worse off than under a report that disclosed the local information.

This is directly analogous to the deworming example in Andrews and Shapiro (2021): censoring negative treatment effect estimates discards information that some audience member could use. Reporting only the global rank–rank slope when copulas cross discards information about *where* concordance fails—information that some audience member needs.

Local slopes as neutral communication. The rank–rank slope has a constant intensity kernel: $W_\rho(u, v) = 12$ for all (u, v). This makes the local rank–rank slope the most "neutral" local report in the following sense. For any audience member with kernel W_m, the local slope in bin B_j captures the unweighted average of ΔC in that bin:

$$\Delta\rho_j \propto \int_{B_j} \int_0^1 \Delta C(u, v)\, du\, dv. \tag{G.1}$$

An audience member with smooth kernel W_m can approximate their measure's local contribution as

$$\int_{B_j} \int_0^1 W_m(u, v)\, \Delta C(u, v)\, du\, dv \approx \bar{W}_{m,j} \cdot \Delta\rho_j\,, \tag{G.2}$$

where $\bar{W}_{m,j}$ is the average of W_m within bin j. The approximation error depends on the variation of W_m within the bin and vanishes as the partition refines. Because the local slope itself uses uniform weights, this approximation is equally good for all smooth-kernel audience members—no particular reader is systematically favoured or disfavoured by the choice of local statistic.

For audience members with concentrated kernels—readers who care about a specific transition probability—the local slope may be insufficient because it averages over the point they care about. Supplementing with transition probabilities at standard thresholds (step (c) of the protocol) addresses this directly. The combination of local slopes and selected transition probabilities is thus approximately communication-admissible for the class of concordance-monotone measures encompassing both smooth and concentrated intensities.

Minimum partition size. The Lipschitz bounds in Appendix F provide a formal criterion for the coarseness of the reporting partition. For a measure with Lipschitz constant L_M, the error from using a J-bin partition rather than the full ΔC surface is of order L_M/J. A partition is communication-admissible for the class of measures with Lipschitz constant bounded by $\bar{L}$ when $\bar{L}/J$ is small relative to estimation uncertainty. For the standard measures in Table 1, this is typically achieved with a small number of regions.

H. Additional Results for Section 7: Welfare-Consistent Completions

An alternative approach to completing the concordance order incorporates welfare judgments. A planner ranks economies based on distributional preferences over parent–child outcomes.

Reinterpreting equation (15), the planner's choice of measure implicitly assigns welfare weights $\phi_M(Y^K, Y^P)$. When $\phi_M(Y^K, Y^P)$ is submodular, the planner has a preference for a decrease in concordance. This completion respects concordance rankings while breaking ties via planner preferences. It accommodates utilitarian welfare with concave utility over dynastic income, opportunity-equality objectives minimising rank correlation, and poverty-focused weights concentrating on low-rank dynasties. This extends results in Atkinson and Bourguignon (1982) to all submodular social welfare functions.

I. Construction of the Replication Dataset

The construction of the data follows Jácome, Kuziemko, and Naidu (2025). I briefly summarize the key features of their approach here and refer the interested reader to Jácome, Kuziemko, and Naidu (2025).

Data sources. I pool the fifteen U.S. surveys identified by Jácome, Kuziemko, and Naidu (2025) that report (i) respondents' current family income, (ii) father's occupation when the respondent was growing up, (iii) respondent race, and (iv) birthplace or childhood region (South vs. non-South). The surveys include American National Election Studies, General Social Survey, the Panel Study of Income Dynamics (specifically 1997 and 2017 waves), Occupational Changes in a Generation 1962 and 1973 surveys, the National Longitudinal Surveys, and others listed in Appendix E of their paper.

I use IPUMS census extracts from 1910 to 2019 (American Community Survey). These extracts differ from the original analysis which uses the full 1940 census available on the NBER server. They may also differ in other years due to differences in the size of the census-extracts selected.

Sample. I retain U.S.-born men aged 30–50, the window that best approximates permanent income while minimising life-cycle bias. Respondents must have non-missing family income, race, region, and father's occupation. Foreign-born individuals are excluded because parental incomes are imputed from U.S. sources.

Respondent income. Each survey's family-income question is harmonised into 10–12 real-1950-dollar bins; continuous responses are recoded to the bin mid-point. Bottom-coded observations (including 0) are set to 0.75 × the upper boundary of this lowest bin and those in the top-bin are set to 1.25 × lower-bin boundary.

Parental income imputation. Fathers' occupations are mapped into 28 broad categories (e.g., skilled crafts, farm operators). Predicted parental family income is the mean household income in matching occupation × race × South cells drawn from historical micro-data: 1901 Cost of Living Survey & 1900 Census (early cohorts), full-count 1940 Census plus the 1936 Expenditure Survey, and 1960–1990 Censuses. For farmers and self-employed, incomes are adjusted following Collins and Wanamaker (2022); non-working fathers receive values imputed from contemporaneous Census means. Father race is proxied by respondent race; father region by respondent childhood region.

Weighting. Where surveys supply sampling weights, I re-centre them to mean 1. I then adjust all surveys so that each birth-decade cell matches Census race population shares (white men and Black men); surveys lacking weights receive weight equal to 1 before adjustment.

Cohorts. Decade birth cohorts 1910s–1970s are used. For each respondent (a) log family income and (b) rank in their cohort specific income distribution are computed; parental income is treated analogously, yielding comparable distributions of income ranks across cohorts.

These steps replicate the long-run, nationally representative parent–child dataset used in the original study and permit the replication of Figure 1 in their paper (Panel (A), Figure 7 here).

J. A Testing Procedure for the Concordance Order

I evaluate the concordance ordering condition on a finite grid $\mathcal{G} = \{(u_g, v_g)\}_{g=1}^{G}$ with $u_g = j_1/J$ and $v_g = j_2/K$ for $j_1 = 1, \ldots, J-1$ and $j_2 = 1, \ldots, K-1$, giving $G = (J-1)(K-1)$ interior grid points. With a fixed grid, the procedure detects alternatives that violate the concordance inequality at an evaluated point; a finer grid increases power against more localized violations at the cost of computation.[35] Define

$$m_g = C^A(u_g, v_g) - C^B(u_g, v_g), \qquad g = 1, \ldots, G. \tag{J.1}$$

The grid-restricted null is

$$H_0 : m_g \geq 0 \ \text{ for all } g = 1, \ldots, G \qquad \text{vs.} \qquad H_1 : m_g < 0 \ \text{ for some } g. \tag{J.2}$$

These are finite-dimensional inequality restrictions on copula functionals. Unlike standard moment inequalities written directly as sample averages of observed variables, the feasible copula estimator also depends on estimated marginal distributions through the rank transformation. The empirical checkerboard-copula process below accounts for this first-stage estimation. I use $J = K = 20$ (giving $G = 361$).

Let $\{(Y_i^{K,s}, Y_i^{P,s})\}_{i=1}^{n_s}$ be independent random samples from population $s \in \{A, B\}$, with the two samples independent of each other. Let $\widehat{F}^s$, $\widehat{F}_K^s$, and $\widehat{F}_P^s$ denote the empirical joint and marginal distributions for group $s \in \{A, B\}$. I construct the empirical checkerboard copula $\widehat{C}_s$ as the multilinear extension of the empirical subcopula on the ranges of $\widehat{F}_{K,s}$ and $\widehat{F}_{P,s}$, following the convention in Section 2. This construction averages over the random tie-breaking device within each mass point and therefore does not depend on a particular realization of jittered ranks. The estimated inequality at grid point g is

$$\widehat{m}_g = \widehat{C}^A(u_g, v_g) - \widehat{C}^B(u_g, v_g). \tag{J.3}$$

[35] This is a standard approach in the univariate stochastic dominance setting, but it is consistent only against alternatives that violate $C^A \geq C^B$ at the fixed grid point. A finer grid increases power against localized alternatives at the cost of computation. Intuitively, the granularity of the grid should increase as the sample size increases. As the variance of comparisons in the tails can be very large, it also serves to effectively truncate the tails.

The key distributional result is the weak convergence of the empirical multilinear copula process. Genest, Nešlehová, and Rémillard (2017) establish this result under broad conditions allowing arbitrary marginal distributions. Their regularity condition requires continuous first-order partial derivatives of the checkerboard copula on an open subset of the unit square and is automatically satisfied when the underlying margins are discrete. Thus, for each cohort,

$$\sqrt{n_s}\left(\widehat{C}^s - C^s\right) \;\to^d\; \mathbb{G}^s \tag{J.4}$$

Evaluating the process at the finite set of grid points yields joint asymptotic normality of $\widehat{m} = (\widehat{m}_1, \ldots, \widehat{m}_G)'$. I write the resulting limit as

$$\sqrt{n}(\widehat{m} - m) \;\to^d\; N(0, \Omega),$$

where Ω is the covariance matrix of the joint limiting vector and n denotes the effective sample-size normalization. When the cohort samples and all auxiliary quantities are independent across cohorts, Ω has the corresponding block-independent form; the bootstrap implementation below allows the covariance structure to be estimated directly.

J.1. Test Statistic

The test statistic is the minimum of the studentized inequalities:

$$T_n = \min_{g=1,\ldots,G} \frac{\widehat{m}_g}{\widehat{s}_g}, \tag{J.5}$$

where $\widehat{s}_g$ is the bootstrap standard error of $\widehat{m}_g$. In each replication I reconstruct the empirical marginal and joint distributions and recompute the checkerboard copula, so that the bootstrap incorporates the sampling variation generated by the estimated rank transformation and the observed mass points. Under H_0, $m_g \geq 0$ for all g, so large negative values of T_n provide evidence against the null. If all inequalities are strictly slack, T_n diverges to $+\infty$. I adopt the generalized moment selection (GMS) procedure of Andrews and Soares (2010) (see also Andrews and Barwick 2012) to determine which inequalities are near-binding and which are clearly slack. The empirical checkerboard-copula result above supplies the finite-dimensional Gaussian approximation for $\widehat{m}$ and the bootstrap approximation to its sampling distribution; GMS determines which components of this vector enter the critical value. Following Andrews and Soares (2010),

I set $\kappa_n = \sqrt{2 \log \log n}$, where $n = n_A + n_B$. This sequence diverges slowly, so truly slack inequalities are eventually excluded while near-binding inequalities remain in the reference distribution.

Relative to the intersection–union min-t procedures used in the concordance-order literature (e.g., Denuit and Scaillet 2004), the GMS step reduces conservativeness when many of the evaluated inequalities are slack.

Continuous margins. When both marginal distributions are continuous, the checkerboard construction reduces to the ordinary empirical copula and the standard empirical-copula process results apply (e.g., Fermanian, Radulovic, and Wegkamp 2004). Thus, the discrete-margin treatment above contains the continuous case as a special case.

J.2. Two-Sided Assessment

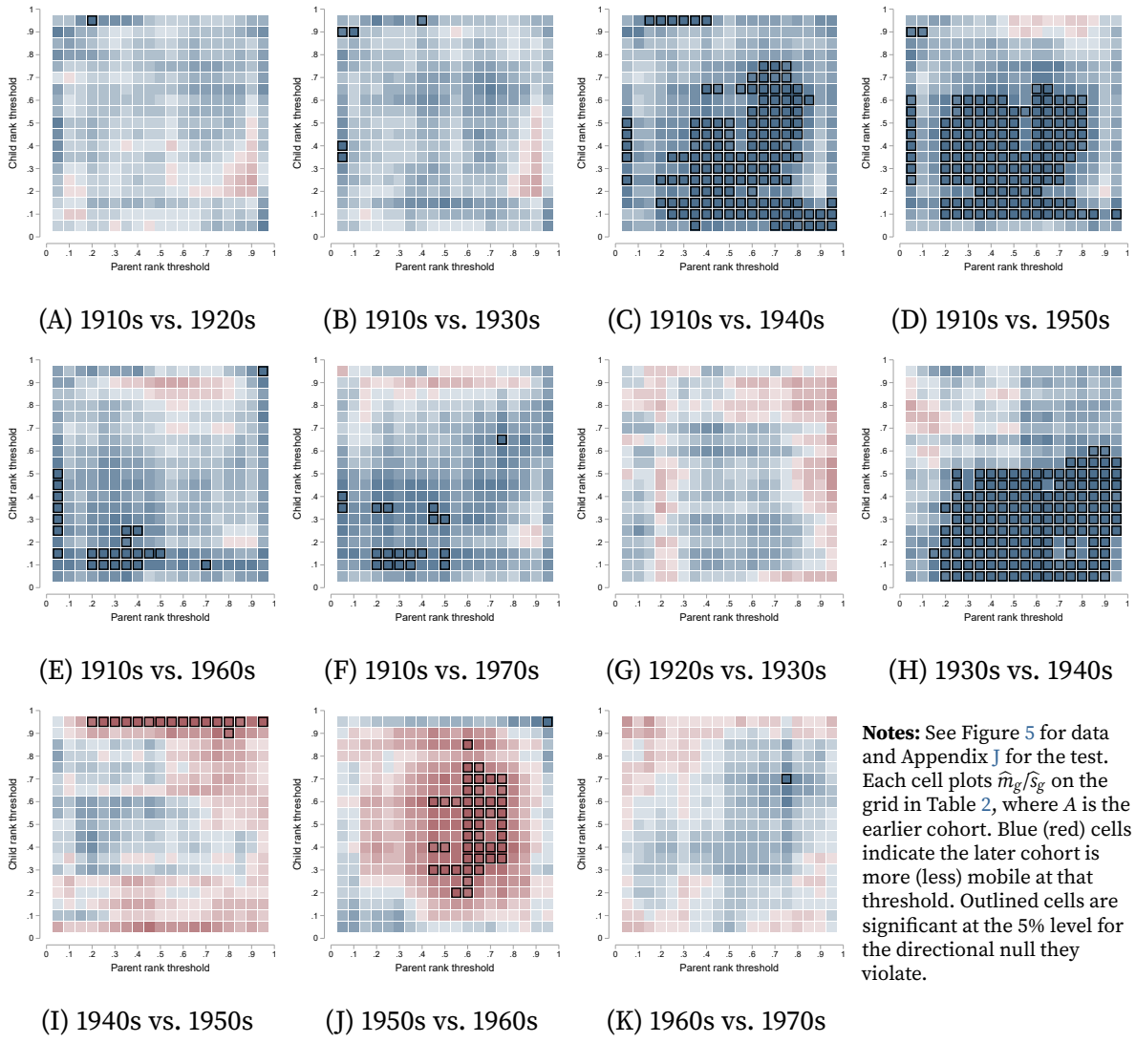
To assess whether the copulas can be ranked, I run the test in both directions: first testing $H_0 : C^A \succeq C^B$, then $H_0 : C^B \succeq C^A$. Four outcomes are possible:

Reject $C^A \succeq C^B$?	Reject $C^B \succeq C^A$?	Test outcome
No	No	Reject neither
Yes	No	Reject $C^A \succeq C^B$ only
No	Yes	Reject $C^B \succeq C^A$ only
Yes	Yes	Reject both

Failure to reject a directional null does not establish the corresponding population ordering. I therefore report which directional null hypotheses are rejected at the 5% significance level, without interpreting non-rejection as affirmative evidence of concordance.

K. Additional Results for Section 8

Figure K.4 repeats the exercise of Section 8 for two additional axiomatic indices. Panel (A) plots the mobility measure of Fields and Ok with exchange (left axis) and total (right axis) mobility; Panel (B) shows the Cowell and Flachaire measure with "relative status" computed using income ranks and levels. Both rise across successive birth cohorts, echoing the upward trends in rank–based measures from Figure 7, even though the

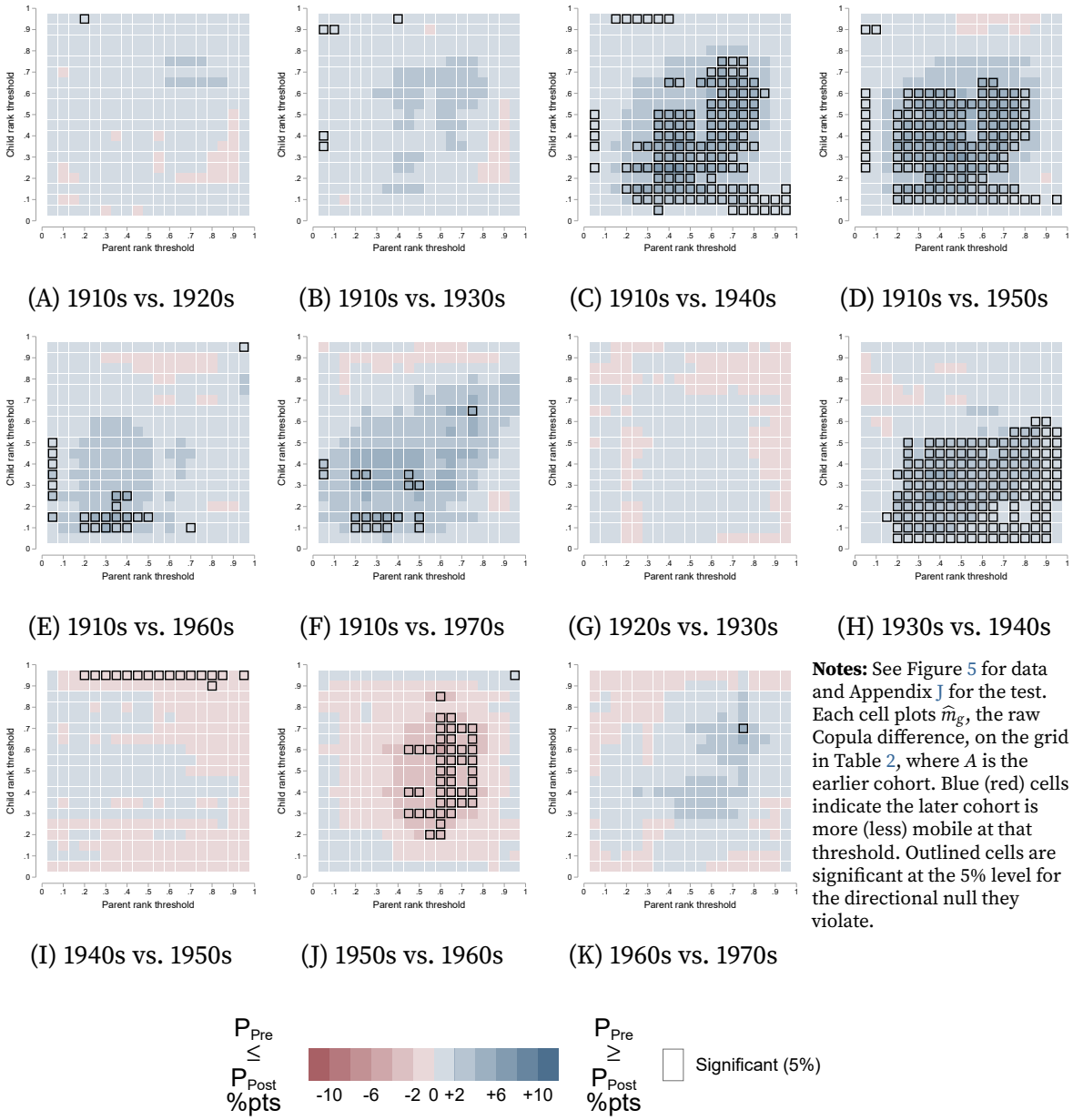


Notes: See Figure 5 for data and Appendix J for the test. Each cell plots $\widehat{m}_g/\widehat{s}_g$ on the grid in Table 2, where A is the earlier cohort. Blue (red) cells indicate the later cohort is more (less) mobile at that threshold. Outlined cells are significant at the 5% level for the directional null they violate.

FIGURE K.1. Locating Concordance Violations Between Birth Cohorts

concordance order does not, by itself, determine the behaviour of these measures when marginals differ across cohorts.

By contrast, the absolute version of the Fields and Ok index, which incorporates growth in mean income and changes in inequality, tells a different story. It climbs sharply for early-century cohorts but flattens thereafter, much like the reversal in the IGE. Aggregate income growth concentrated in the right tail of the income distribution (see, e.g., Blundell et al. 2018; Piketty, Saez, and Zucman 2018; Guvenen et al. 2022)

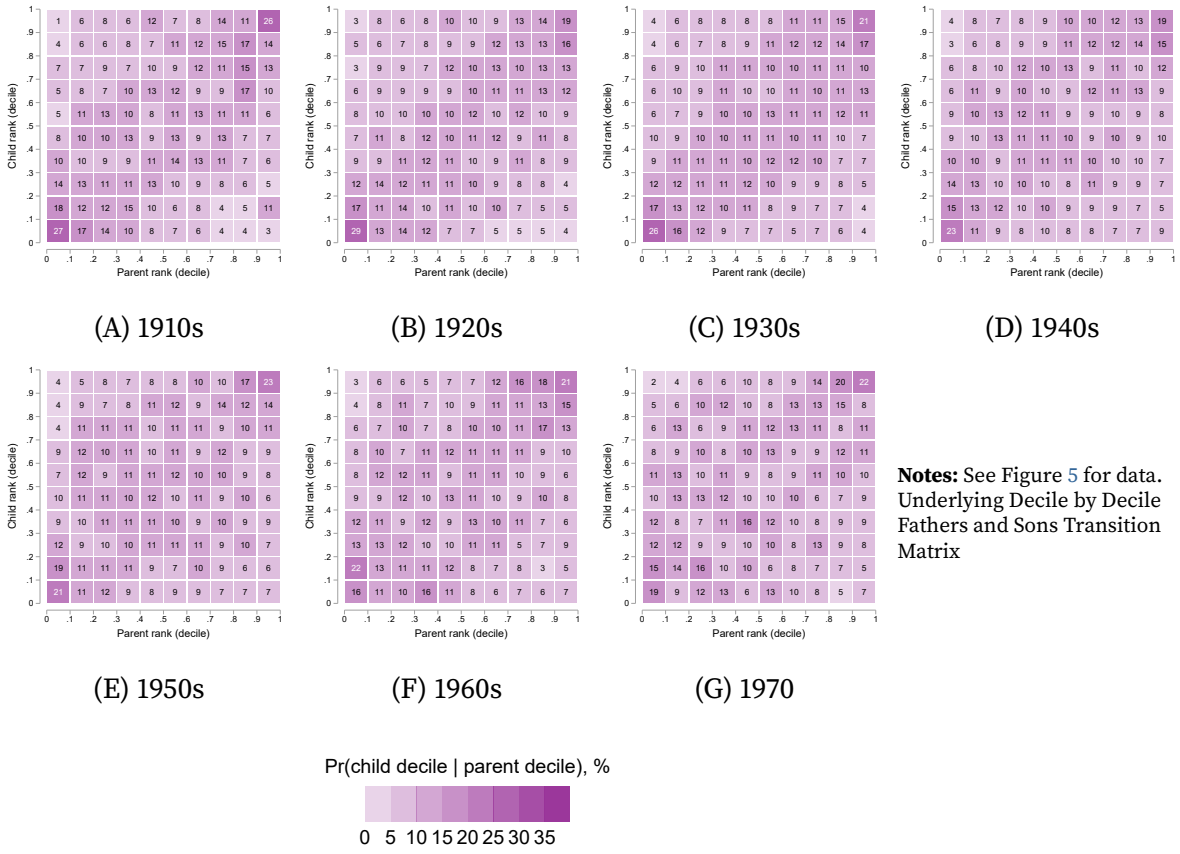


Notes: See Figure 5 for data and Appendix J for the test. Each cell plots $\widehat{m}_g$, the raw Copula difference, on the grid in Table 2, where A is the earlier cohort. Blue (red) cells indicate the later cohort is more (less) mobile at that threshold. Outlined cells are significant at the 5% level for the directional null they violate.

FIGURE K.2. Raw Copula Deviations Between Birth Cohorts

offsets the gains in relative mobility, leaving absolute mobility roughly constant in the latter half of the century.

The decomposition in equation (11) allows me to quantify how much of the change in each measure reflects shifts in the dependence structure versus changes in marginal distributions. Figure K.4 reports results using Shapley-Owen-Shorrocks averaging over both orderings. This answers why level–based measures diverge from rank–based



Notes: See Figure 5 for data. Underlying Decile by Decile Fathers and Sons Transition Matrix

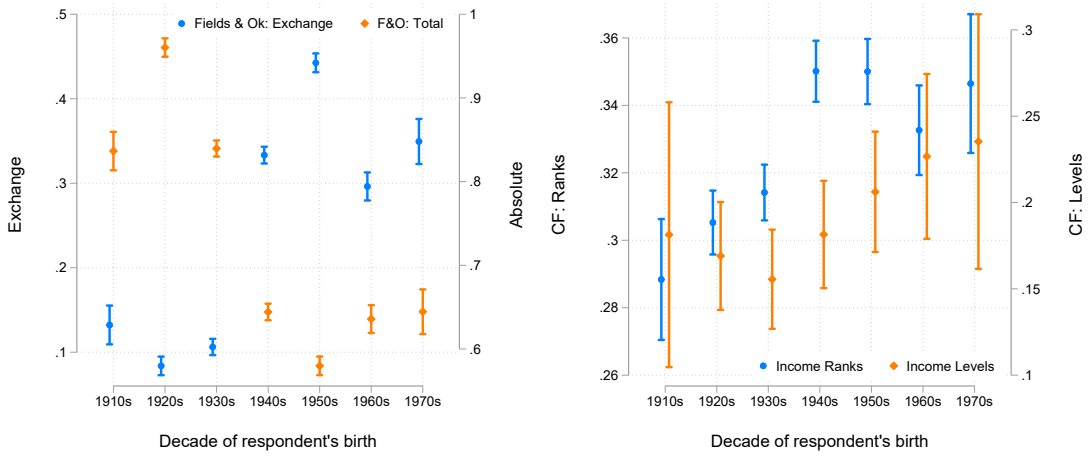
FIGURE K.3. Birth Cohort Transition Matrices

measures in practice.

Panels (C) and (D) show that, like the measures reported in the main text, Fields and Ok measures similarly decompose into dependence-driven gains and marginal changes. In relative units, the decomposition in equation (11) reveals that these are driven mainly by changing marginals. In this application, the mobility measures that depend on the level of (log-)income are both highly sensitive to and dominated by the evolution of the income distribution over the 20th century.

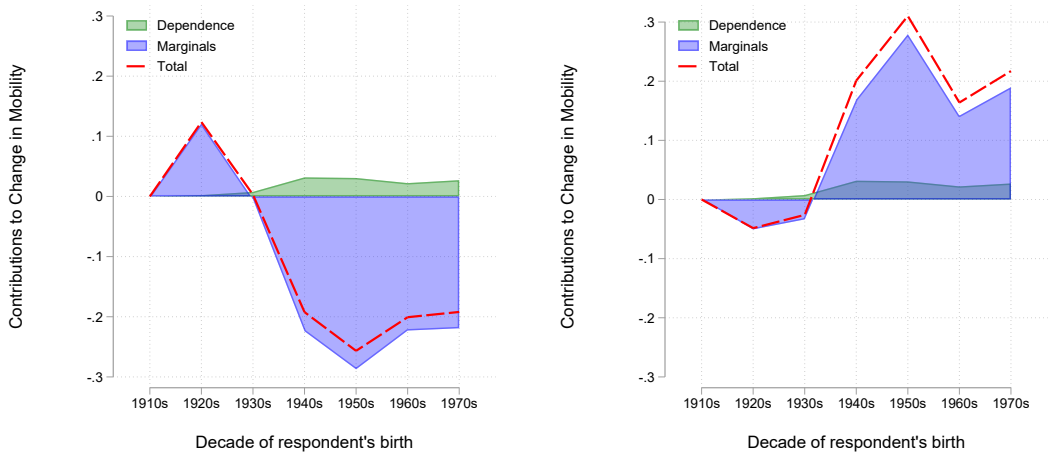
L. Proofs for Section 5

Proof of Proposition 6. Let $\varepsilon = \ln \theta$ and $X^s = S^s(Y^P) + \varepsilon$. Since W^s is strictly increasing, $(Y^{K,s}, Y^P)$ and (X^s, Y^P) share a copula. For $z \in [0,1]$, let $S_z = S^B + z\Delta$ and $X_z = S_z(Y^P) + \varepsilon$ with copula C_z, so that $C_0 = C^B$ and $C_1 = C^A$.



(A) Fields & Ok Measure

(B) Cowell & Flachaire Measure



(C) Fields & Ok (Total)

(D) Fields & Ok (Transition)

FIGURE K.4. Trends in Intergenerational Mobility Measures (cont.)

Notes: See Figure 5 for data construction details. Panels (A) and (B) report estimates of the Fields & Ok and Cowell & Flachaire measures (Proposition 4), respectively. Panels (C) and (D) report Fields and Ok (1996, 1999) total mobility and exchange (transition) mobility measures, respectively. Each panel decomposes the change in mobility (relative to the 1910s cohort) into a dependence (copula) component and a marginal component following equation (11), with Shapley averaging over both orderings to address path dependence.

Fix $u \in (0,1)$. Because $f_\varepsilon > 0$, the marginal CDF of X_z is strictly increasing, and

$$u = \mathbb{E}\left[F_\varepsilon\left(q_z - S_z(Y^P)\right)\right]$$

defines a unique quantile q_z. Boundedness of S^s and of f_ε permits differentiation under

the expectation, giving

$$\dot{q}_z = \frac{\mathbb{E}[w_z \Delta]}{\mathbb{E}[w_z]}, \qquad w_z(y) = f_\varepsilon\big(q_z - S_z(y)\big) > 0.$$

The weight $w_z(y)$ is the density of X_z at q_z conditional on $Y^P = y$, so $\dot{q}_z = \mathbb{E}[\Delta(Y^P) \mid X_z = q_z]$: the quantile moves with the average incidence among the parents of children at rank u.

Fix $v \in (0,1)$, and let $L = \{Y^P \le F_P^{-1}(v)\}$ and $H = L^c$. Since $C_z(u,v) = \mathbb{E}[\mathbf{1}_L F_\varepsilon(q_z - S_z(Y^P))]$,

$$\frac{\partial C_z(u,v)}{\partial z} = \mathbb{E}\big[\mathbf{1}_L w_z(\dot{q}_z - \Delta)\big] = \frac{W_L W_H}{W_L + W_H}\left(\bar{\Delta}_H - \bar{\Delta}_L\right), \tag{L.1}$$

where $W_J = \mathbb{E}[w_z \mathbf{1}_J] > 0$ and $\bar{\Delta}_J = \mathbb{E}[w_z \Delta \mathbf{1}_J]/W_J$ for $J \in \{L, H\}$. If Δ is nondecreasing, every parental resource level in H exceeds every level in L, so $\bar{\Delta}_H \ge \bar{\Delta}_L$. Hence $C_z(u,v)$ is nondecreasing in z at every (u,v), and $C^A \succeq C^B$ since copula boundary values coincide. The nonincreasing case reverses the inequalities.

For part (ii), replacing Δ by $\Delta + k$ for a constant k replaces X^A by $X^A + k$, a common shift that leaves every child's rank, and therefore C^A, unchanged. Hence C^A depends on Δ only through $\Delta - \mathbb{E}[\Delta(Y^P)]$; if Δ is constant, X^A and X^B differ by a constant and $C^A = C^B$. For part (iii), given f^s and $I^{\star s}$, X^s does not depend on W^s, and $Y^{K,s} = W^s\big(\exp(X^s)\big)$ is a strictly increasing transformation of X^s, so the copula of $(Y^{K,s}, Y^P)$ is that of (X^s, Y^P) for every strictly increasing W^s. If $\ln \theta^s = \sigma^s \varepsilon$, apply the argument to X^s/σ^s, which has the same ranks, with $\Delta = S^A/\sigma^A - S^B/\sigma^B$. □

Proof of Corollary 3. With log utility, the first-order condition $1/[y + T^s(y) - I] = \delta^s \kappa^s \alpha^s / I$ and strict concavity give $I^{\star s}(y) = c^s[y + T^s(y)]$. Hence

$$S^s(y) = \alpha^s \ln c^s + \alpha^s \ln\big(y + T^s(y)\big) + \gamma^s \ln y.$$

In part (i), $\Delta(y) = \Delta_0 + (\alpha^A - \alpha^B + \gamma^A - \gamma^B) \ln y$ for a constant Δ_0, which is nondecreasing. In part (ii), $\Delta = \alpha \ln\left(c^A(1+\tau)/c^B\right)$ is constant, and c^s is increasing in $\delta^s \kappa^s$. Since $\ln Y^{K,s} = \ln \omega^s + \kappa^s[S^s(Y^P) + \ln \theta]$, the IGE is $\beta^s = \kappa^s(\alpha + \gamma)$. In part (iii), $\Delta(y) = \alpha \ln(1 + t/y)$ is decreasing, while $c(y + t) > c\,y$. The orderings follow from Proposition 6. □

Proof of Corollary 4. I prove a sharper statement. Let $\bar{\Delta}(u) = \mathbb{E}[\Delta(Y^P) \mid X^B = q^B(u)]$ denote the average gain among the parents of children at rank u in economy B. It suffices that $\min\{\Delta(\underline{y}), \Delta(\overline{y})\} > \bar{\Delta}(u)$ for some $u \in (0,1)$. The condition in the main text

implies this: since $F_{X^B}(X^B)$ is uniform, the law of iterated expectations gives $\int_0^1 \bar{\Delta}(u)\,du = \mathbb{E}[\Delta(Y^P)]$, so $\bar{\Delta}(u) \leq \mathbb{E}[\Delta(Y^P)]$ for some u.

Fix such a u and evaluate equation (L.1) at $z = 0$, with $w_0(y) = f_\varepsilon(q^B(u) - S^B(y))$, which is bounded above and away from zero on the compact support of parental resources, so that $\bar{\Delta}(u) = \mathbb{E}[w_0\Delta]/\mathbb{E}[w_0]$. As $\nu \to 0$, continuity of Δ gives $\bar{\Delta}_H - \bar{\Delta}_L \to \bar{\Delta}(u) - \Delta(\underline{y}) < 0$. As $\nu \to 1$, it gives $\bar{\Delta}_H - \bar{\Delta}_L \to \Delta(\overline{y}) - \bar{\Delta}(u) > 0$. There therefore exist ν_1 and ν_2 at which the derivative at $z = 0$ is strictly negative and strictly positive, respectively. By differentiability, $C_z(u, \nu_1) < C^B(u, \nu_1)$ and $C_z(u, \nu_2) > C^B(u, \nu_2)$ for all sufficiently small $z > 0$, so neither economy dominates the other. □

A multi-period technology. Suppose log skill accumulates as

$$q_{t+1} = a_t + \pi_t q_t + \sum_j \phi_{jt} \ln z^s_{jt}(y) + \xi_t,$$

where z^s_{jt} are input policy functions for goods or parental time. Iterating, with $\Pi_t = \prod_{r=t+1}^{T} \pi_r$, and holding the initial-skill gradient and coefficients common across economies, incidence is

$$\Delta(y) = \Delta_0 + \sum_{t,j} \Pi_t \phi_{jt} \ln \frac{z^A_{jt}(y)}{z^B_{jt}(y)}.$$

Proposition 6 applies whenever the aggregate shock $\sum_t \Pi_t \xi_t$ satisfies its conditions. The relevant condition is that the weighted proportional change in inputs is monotone in parental resources, not that every input rises.

When the reform is indexed by a continuous parameter λ, such as the log price of an input or a subsidy rate, write $z_{jt}(y;\lambda)$. Each log ratio is then a cumulated Marshallian elasticity of input demand,

$$\ln \frac{z^A_{jt}(y)}{z^B_{jt}(y)} = \int_{\lambda^B}^{\lambda^A} \varepsilon_{jt}(y;\lambda)\,d\lambda, \qquad \varepsilon_{jt}(y;\lambda) \equiv \frac{\partial \ln z_{jt}(y;\lambda)}{\partial \lambda},$$

and incidence cumulates the elasticity of child skill with respect to the reform, $\partial S(y;\lambda)/\partial\lambda = \sum_{t,j} \Pi_t \phi_{jt} \varepsilon_{jt}(y;\lambda)$. It is therefore sufficient that this skill elasticity is monotone in parental resources at every λ along the reform path—that is, that log skill has increasing (or decreasing) differences in (y, λ). A subsidy can lower concordance

either because poorer families substitute more toward the subsidised input or because it relaxes their budgets by more. In richer models such as Del Boca, Flinn, and Wiswall (2014), input policies respond to wages and non-labour income jointly through labour supply, so the relevant elasticities are defined with respect to full income and checking the condition requires their estimated policy functions.

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